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The Role of Mobile Communications for Industrial Automation: Architecture, Applications and Challenges

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ABSTRACT Industrial automation is expected to benefit from the capabilities of Beyond 5G (B5G)/6G, particularly in improving real-time decision making and control in information-driven scenarios, supporting improved efficiency, precision and safety in connected manufacturing environments. While not all aspects of industrial automation are equally affected, certain areas such as real-time monitoring, wireless control and predictive maintenance can benefit significantly from the high reliability, ultra-low latency and scalability of next-generation wireless systems. Motivated by the increasing convergence between operational technology (OT) and advanced communication infrastructures, this paper surveys how B5G/6G can serve as an enabler for data-centric industrial ecosystems. The main contribution lies in analysing evolving industrial architectures, the integration of OT systems and application domains including digital twins, cloud robotics and smart devices. At the end of the paper, we also offer strategic insights into enabling technologies and future challenges, providing a comprehensive overview of how B5G/6G can support the evolution of Industry 5.0.

INDEX TERMS Architecture, industrial automation, B5G, 6G, applications.

I. INTRODUCTION

INDUSTRIAL automation refers to the use of technology and machines to streamline and optimize industrial processes [1]. Industry 4.0 is on the rise as manufacturing processes undergo a digital transformation that aims to connect the physical and digital worlds through the integration of cyber-physical systems [2]. As part of this evolution, Industry 5.0 has started to take shape [3], with a focus on sustainability, resilience, and a human-centered approach [4]. The advancements in wireless technology are

designed to create intelligent connections between humans and machines, bridging the physical world with their digital counterparts in a virtualized digital realm [5], [6]. The progress in cloud computing [7], and edge computing [8], artificial intelligence, and robotics unlock new opportunities to revolutionize these processes, opening up new dimensions of possibilities [9]. In addition, the concept of digital twinning allows for the support of engineering, maintenance, reconfiguration, and recycling of industrial automation systems [10]. This opens up possibilities for

various visionary use cases that will demand enhanced communication capabilities such as found in B5G/6G.

The provision of ultra-Reliable Low-Latency Communication (uRLLC) has been a crucial element of 5G technology [11]. This focus on reliable and low-latency communication will continue to evolve within the context of the 6G vision. While the global Industry 4.0/5.0 market is anticipated to experience significant growth by 2028, understanding the system architecture, applications, and challenges that these industries pose for wireless systems is crucial. The utilization of 6G communication will not only facilitate but also drive advancements in various industry verticals, thereby playing a pivotal role in enhancing system efficiency and availability [12]. Hence, there is a need for a comprehensive survey across all supporting technologies. B5G/6G can be used to track and optimize production and maintenance processes, enable autonomous production, and provide predictive maintenance, resulting in lower costs and improved product quality. B5G/6G could also enable intelligent robotic systems or brain-computer interfaces that can interact and collaborate with humans, resulting in improved production and safety.

Smart factories have been on the rise due to various technological improvements in recent decades. However, connectivity has been a significant issue. The emergence of 5G has sparked the fourth industrial revolution, introducing several advanced technologies. Industry 5.0 represents the upcoming phase of the industrial revolution, which builds upon the principles of Industry 4.0. Its primary goal is to achieve high levels of accuracy, while simultaneously reducing waste and cost through mass customization. It embodies a vision of networked systems and advanced technologies that will redefine the way industry works [13]. The motivation for this transformation is to harness the potential of cutting-edge technologies to increase efficiency, productivity and safety in unprecedented ways. The selective adoption of 6G technologies can support the evolution toward Industry 5.0 by facilitating joint communication-control co-design, improved context-awareness, and the orchestration of time-critical industrial tasks [14]. This shift is not universal, but focused on specific use cases that demand extreme reliability and low latency. Various verticals, such as healthcare, warehouse and transport sectors will see the extensive use of mobile robot swarms and drones with the widespread adoption of 6G.

A. RESEARCH METHODOLOGY

This survey is conducted to synthesize and analyze the scattered literature on the impact of emerging communication technologies in industrial automation. By consolidating diverse perspectives and findings, a comprehensive understanding of how these technologies can reshape industrial landscapes, addressing both the technological and practical implications is provided. The following research questions guide this survey:

- What impact will advanced wireless technologies such as 5G, B5G and 6G have on industrial architecture and how will they facilitate the transition from Industry 4.0 to Industry 5.0? The answer to this question will be covered in Section II, “Industrial Architecture: Evolution and Convergence of OT with Wireless”.
- How do various industrial applications benefit from the implementation of state-of-the-art communication technologies and what challenges do they pose? The answer to this question will be covered in Section III, “Industrial Applications”.
- How do specific wireless technologies and innovations enable improved efficiency and functionality within industrial applications? The answer to this question will be covered in Section IV “Industrial Applications and Use Cases”.
- What are the key technological enablers that support the deployment and effectiveness of these applications? The answer to this question will be covered in Section V, “Industrial Devices”
- What are the current standardization efforts and important research projects that support the use of 5G and beyond technologies in the industrial environment? The answer to this question will be covered in Section VI, “Standardization and Research Projects”.
- What are the open challenges in integrating Industry 5.0 with advanced wireless communication technologies and what strategies could address these challenges? The answer to this question will be covered in Section VII, “Open Challenges”.
- What lessons have been learned from current implementations of 5G and beyond in the industry, and what future directions exist for research and development? The answer to this question will be covered in Section X, “Lessons Learned and Future Directions”.

All references were selected based on the following criteria: (i) *Relevance*: Publications directly addressing the integration of advanced communication technologies with industrial automation. *Recency*: Emphasis on studies and reviews published within the last five years to ensure the inclusion of the latest advancements and insights. *Credibility*: Preference for peer-reviewed journals, reputable conferences, and authoritative industry reports to ensure the reliability of the information. In addition to Relevance, Recency, and Credibility, technical criteria were also employed to assess the selected references. These include *technical novelty*, *applicability to industrial use cases*, *scalability and performance metrics*, and *evidence of multidisciplinary collaboration*. Each reference was evaluated for its contribution to addressing specific challenges in industrial automation, such as latency, reliability, and integration with emerging Industry 5.0 technologies.

B. RELATED SURVEYS

The integration of B5G/6G technology holds the promise of intelligent systems, with robots in industrial settings to be

capable of interacting and collaborating with their human counterparts seamlessly. The synergy between human ingenuity and robotic precision ushers in a new production era, one characterized by improved safety, heightened productivity, and limitless possibilities. Spinelli and Mancuso [15], explain the applicability of Multi-access Edge Computing (MEC) for the industry verticals, including automotive, smart city, eHealthcare, Augmented Reality (AR)/Virtual Reality (VR) applications and smart factories. However, the article's contribution lacks a detailed analysis of distinguishing architectural capabilities of 5G and beyond networks for industrial automation. Sutton et al. [16], present a comprehensive discussion on the networks' physical and Medium Access Control (MAC) layer with their improved reliability and latency of applications. Even though the authors reflected in-depth technical insights into the physical and MAC layer techniques that yield reliability and reduced latency, the adaptation potential of the capabilities towards industrial automation, a major beneficiary domain, has not been widely explored. Porambage et al. [17], present a comprehensive review on the potential of MEC for Internet of Things (IoT) realization, including the technical aspects and insights of the distinguishing 5G integration technologies such as Network Functions Virtualization (NFV), Software Defined Networking (SDN) and network slicing. However, the contribution does not highlight the strengths of next-generation mobile communication technologies for advancing industrial applications through automation, which is important to identify to encourage enterprises to improve their business values.

Promwongsa et al. [18], present a comprehensive survey on the potential of tactile Internet, including critical reviews on the architectures, algorithms, and significant limitations of 5G for transmitting skills over the network in human-oriented applications. However, the contribution is limited to human-oriented applications such as remote robotic surgery, autonomous driving, and tele-VR phobia treatment which lacks insights into the strengths of mobile communication for industrial transformation through automation. We identified that it is important to explore the strong potential of mobile communication for different industrial applications of the Industry 5.0, beyond human-oriented applications. Schulz et al. [19], explained the design challenges of time-critical industrial applications such as factory automation, process automation, smart grids, and intelligent transportation systems. However, the contribution of [19], is limited to the lower latency of 5G networks and excludes the distinguishing capabilities of 5G networks, including network slicing with a focus of Industry 4.0. Angueira et al. [20], explain the security challenges in the industry's wireless communication systems, focusing on factory automation. However, the scope of this work is limited to security attacks and countermeasures, which do not reflect the distinguishing capabilities of communication networks for industrial automation. Chi et al. [21], reflect insights on the role of network automation with different

perspectives including topologies, strengths, limitations, and the potentials for Industry 5.0. The scope of the article is limited to discussion on network automation and standardization efforts, without elaboration on emphasizing applicability of network automation for different scenarios of the Industry 5.0. Xiang et al. [22], present a comprehensive review on the key enabling technologies of the Industry 5.0, including a robots and advanced communication technologies in 5G/6G. The authors explained the enablers of the Industry 5.0 in more abstract level, instead of explaining deeper with the relevance of concrete deployment scenarios. Mahmood et al. [23], present review insights on the industrial interests corresponding to large-scale wireless Industrial Internet of Things (IIoT) deployment architecture with a broader analysis on technical strengths of the networks. However, the paper lacks in emphasis on standardization efforts and concrete application scenarios of the IIoT to provide a comprehensive understanding to the readers on the context of industrial automation and the networks. Wu et al. [24], present a comprehensive analysis on network slicing management for different industrial applications, including smart transportation, smart energy and smart factory. However, the scope of the article is limited to network slicing and does not include the prominent capabilities of 5G networks towards industrial automation. Elbayoumi et al. [25], present a comprehensive study on non-orthogonal multiple access to support machine-type communication in ultra-dense networks. However, the article does not reflect insights on the architecture relevance of 5G and beyond networks to Industry 5.0 from the perspectives of deployment options and industrial applications. Finally, Table 1 summarizes the key surveys on the communication for the industrial application domains. In summary, this study advances the discourse on industrial automation and wireless communications by taking an in-depth look at how B5G/6G technologies specifically enable the transition to Industry 5.0, exploring new dimensions such as enhanced interoperability, industrial devices and the practical integration of wireless technologies into industrial applications. Unlike previous works listed in Table 10, our analysis provides a comprehensive and detailed examination of current standardization efforts and future research directions, providing key insights and strategic guidance for the seamless integration of these advanced wireless technologies in the industry.

C. PAPER CONTRIBUTION

This paper uniquely explores the transformative potential of B5G/6G technology within industrial automation, their role in transforming traditional automated systems into intelligent, adaptive production environments, thus providing a unique, forward-looking analysis not extensively covered in existing literature. Unlike prior surveys, this work offers an in-depth analysis of challenges and proposes innovative solutions, including new methodologies and tools to enhance the integration of B5G/6G with Operational Technology (OT)

TABLE 1. Summary of important surveys on communication in the industrial applications.

	Analysis of automation for Industry 5.0	Deployment analysis in Industry 5.0	Analysis of NS for Industry 5.0	Analysis on standardization efforts	Concrete Industry 5.0 application analysis	Analysis on future research directions	Remarks
[15]	—	✓	—	—	—	—	The survey is more focused to the MEC-based approaches and does not provide a comprehensive analysis on architectural and deployment capabilities of mobile communication towards Industry 5.0.
[16]	—	—	—	—	—	—	The contribution is limited to physical and mac layer techniques for low-latency communication.
[17]	—	—	—	—	—	—	The article contribution is limited to the insights of MEC for IoT realization and does not reflect the potentials of industrial automation in a detailed view.
[18]	—	—	—	—	—	—	The article's contribution is limited to tactile internet with human-centred applications. An analysis to the manufacturing-based industry automation was not included in this article.
[19]	—	—	✓	—	—	—	The article contribution is limited to the analysis of low-latency and the associated benefits to the industrial applications. The key features such as NFV, network slicing, and architectural features are not discussed.
[20]	—	—	—	—	—	—	The contribution is focused to the security attacks and countermeasures and lacks in the views of applicability towards industry automation.
[21]	—	—	—	—	—	—	This paper reflects the insights of network automation, which is specifically on automated functionalities of future networks. The scope of this article is limited to the role of automation technologies and lacks emphasis on the role of applications in Industry 4.0.
[22]	—	—	—	—	—	—	This survey provides a comprehensive review on key enabling technologies of the Industry 5.0, including the 5G and beyond networks. However, the article lacks in specific evaluation on the applicability of 5G with application scenarios.
[23]	—	—	—	—	—	—	This survey provides insights from a comprehensive review on beyond 5G visions and architecture in the IIoT. The article lacks in concrete emphasis on the application scenarios of 5G in the IIoT.
[24]	—	—	—	—	—	—	This paper provides insights of intelligent network slicing in smart transportation, smart energy, and smart factory use cases. However, the scope is limited to the context of network slicing and does not reflect the overall significance of the 5G and beyond networks from an architectural perspective towards Industry 4.0.
[25]	—	—	—	—	—	—	The contribution is limited to the NOMA-related capabilities and the article does not reflect insights from the perspective of architectural advancements to unlock the automation potentials through 5G and beyond network integration.
This survey	✓	✓	✓	✓	✓	✓	This survey presents a comprehensive analysis on automation for the Industry 5.0 with deployment analysis, spotlights on leading standardization efforts, and the future research directions. It is worth noting that our survey does not detail the technical aspects of communication and industrial applications. Rather, our work focuses on adaptation strategies and future research.



Comprehensive coverage



Partial coverage



Minimal or no coverage

communication systems. The main contributions of the paper can be summarized as follows:

- The integration of Industry 5.0, a transformative phase in industrial automation, with the breakthrough wireless communication technology for 6G is investigated. Moreover, how these two areas overlap and what potential they offer for industrial applications is also examined.
- An in-depth analysis of industrial architecture, focusing on the integration of OT and its relationship to B5G/6G is provided. It covers key components such as OT integration, the framework for Industry 4.0, the role of wireless communication technologies and a range of deployment options that are important for the seamless operation of industrial systems in the context of industrial automation.
- An overview of industrial devices, the strategies and approaches for implementing this concept in industrial applications (including real-time wireless sensing,

motion control, digital twin concepts, cloud robotics, and machine learning to detect anomalies and assist in troubleshooting) that will be transformed by the integration of 6G technology are investigated.

- In addition to lessons learned and the challenges to be faced, strategic guidelines for the future are also offered by outlining the future directions that will shape the development of Industry 5.0 and the integration of 6G into industrial automation.

D. OUTLINE

The roadmap of the survey is illustrated in Figure 1 and can be summarized as follows: Section II provides a comprehensive overview of the industrial architecture, touching upon the integration of OT, the Industry 4.0 framework, and wireless communication technologies such as 5G. Section III, explores a range of industrial applications and use cases that stand to benefit from the 6G revolution, including wireless real-time sensing, motion control, digital twin concepts,

cloud robotics. Section IV provides the technology enablers for industry automation including machine learning for anomaly detection and troubleshooting, Generative AI and end-to-end resource management. Section V investigates industrial devices, offering an in-depth understanding of its standardization, implementation and real-world applicability. Section VI provides the most important standardization activities and research projects relevant to mobile communication for industrial automation. Section VII categorizes the open challenges related to security and privacy. Section X offers insights into the lessons learned, the challenges faced, and the future directions that will shape the evolving landscape of Industry 5.0 and B5G/6G integration. Section XI concludes the exploration, summarizing the key takeaways and emphasizing the significance of this industrial transformation. Finally, Table 2 provides the notation table of the paper.

II. INDUSTRIAL ARCHITECTURE: EVOLUTION AND CONVERGENCE OF OT WITH WIRELESS

The evolution of operational technology (OT) communication systems is undergoing a pivotal transformation through selective integration with advanced wireless technologies such as 5G, B5G, and 6G networks particularly in scenarios requiring high reliability, real-time coordination, and mobility. This section explores the seamless merging of traditional industrial systems with modern wireless infrastructures and shows how this convergence underpins the transition from Industry 4.0 to the visionary Industry 5.0. Detailed discussions cover the fundamental aspects of OT integration, highlight the transformative impact of time-sensitive networking, and explore the potential of 6G to enhance industrial automation capabilities. Subsequent subsections address the evolution of industry standards, deployment strategies for B5G technologies, and key positioning and timing considerations in this rapidly evolving landscape.

A. OT COMMUNICATION AND 5G/B5G/6G

1) OT INTEGRATION

Industrial communication systems use a range of technologies to support various functions. An automation pyramid, which is a hierarchical communication structure, is frequently employed in the context of manufacturing systems in factories. This structure, shown on the left side of Figure 2 is based on the model ISA 95 [26], and ISA 99 [27]. Within this concept, industrial equipment is connected in smaller subsystems that correspond to specific units, such as a production cell. These subsystems are isolated by gateways and may utilize different communication technologies. Each subsystem is closely managed to ensure reliable and efficient communication performance, especially for critical operations [28]. Moving up the hierarchy, these subsystems are interconnected to enable coordination between production cells and supervisory control of the entire production system. The domain responsible for manufacturing operations, known as the OT domain, encompasses this

TABLE 2. List of acronyms.

Acronym	Definition
3GPP	3rd Generation Partnership Project
4G, 5G, 6G	4th, 5th, 6th Generation
AI	Artificial Intelligence
AR	Augmented Reality
B5G	Beyond 5th Generation
BCN	Blockchain Network
C2C	Controller-to-Controller
CNF	Cloud Native Function
DCN	Dedicated Core Network
DCS	Distributed Control System
eMBB	enhanced Mobile Broadband
ERP	Enterprise Resource Planning
GenAI	Generative AI
IEEE	Institute of Electrical and Electronics Engineers
IIC	Industry IoT Consortium
IIoT	Industrial Internet of Things
IoT	Internet of Things
IMT	International Mobile Telecommunications
ISO	International Organization for Standardization
IRT	Isochronous Real-Time
IT	Information Technology
LCM	Life Cycle Management
LPWAN	Low Power Wide Area Network
MAC	Medium Access Control
MEC	Multi-access Edge Computing
ML	Machine Learning
mMTC	massive Machine-Type Communications
MNO	Mobile Network Operator
NFV	Network Functions Virtualization
NR	New Radio
OPC	Open Platform Communications
OT	Operation Technology
PDN	Packet Data Network
OIA	Open Industry 4.0 Alliance
PLC	Programmable Logic Controller
RAN	Radio Access Network
QoS	Quality of Service
RedCap	Reduced Capability User Equipment
RF	Radio Frequency
SDN	Software Defined Networking
SDO	Standards Developing Organization
SSI	Self Sovereign Identity
STO	Safe Torque Off
SLA	Service Level Agreement
SME	Small and Medium-sized Enterprises
SoC	System on Chip
TSN	Time Sensitive Network
UE	User Equipment
uRLLC	ultra-Reliable Low-Latency Communication
VPN	Virtual Private Networks
QoS	Quality of Service
vRAN	Virtualized Radio Access Network
XR	Extended Reality

interconnected network of subsystems. In the OT domain, the communication requirements tend to be more demanding, particularly at the lower levels. Currently, critical communication primarily relies on wired technologies like fieldbus or industrial Ethernet. It’s important to note that the OT domain is securely segregated from the Information Technology (IT) domain, which encompasses enterprise applications and services. This separation ensures the integrity and isolation of the critical communication within the OT domain.

2) 5G AND THE ROLE OF TIME SENSITIVE NETWORK

Wireless connectivity offers significant benefits to manufacturing systems, including cost savings by reducing the need for extensive cabling, enabling new use cases that involve mobile components, and supporting the overall trend towards Industry 4.0 [29]. The introduction of 5G technology has delivered reliable and deterministic low-latency services,

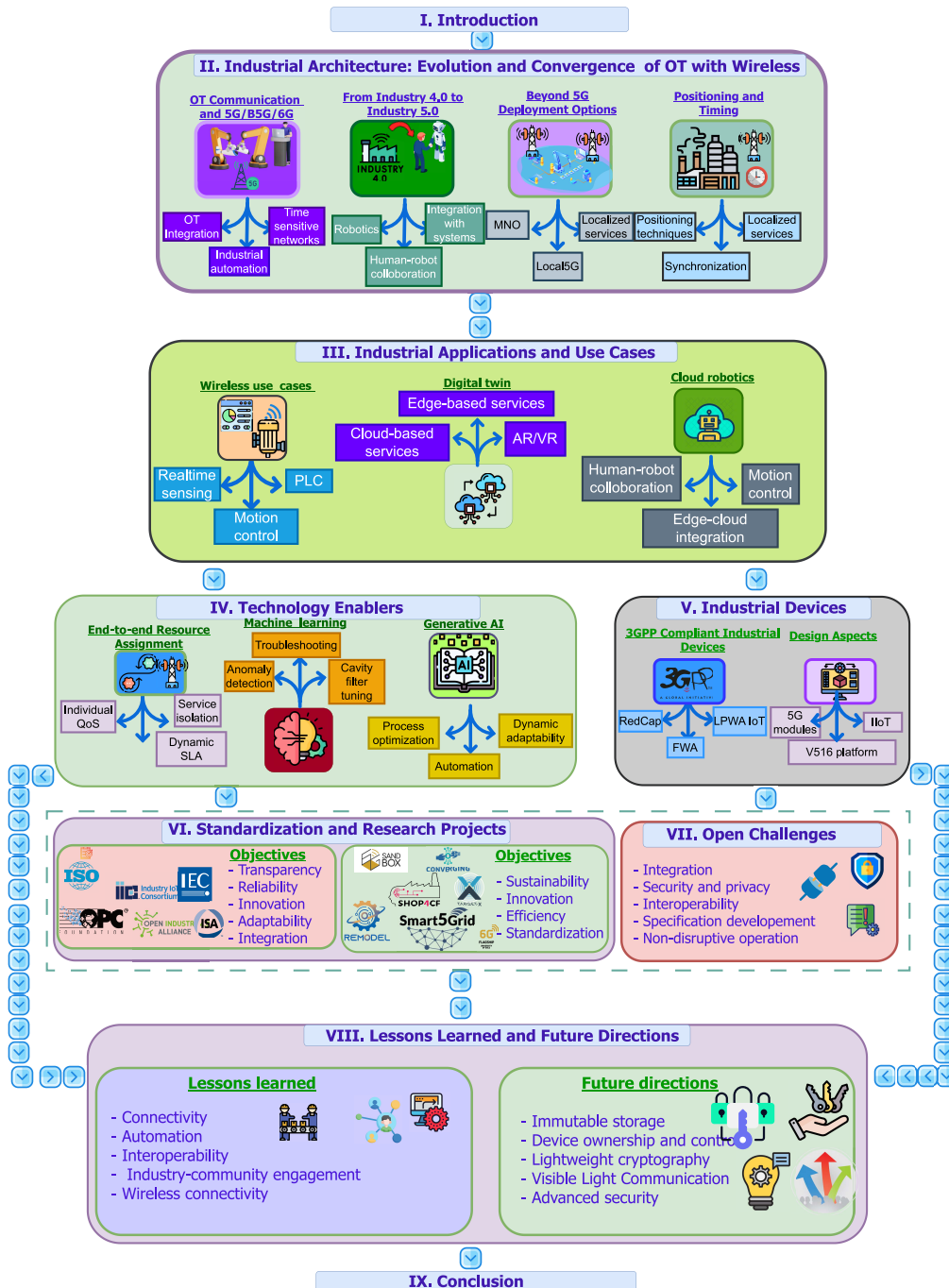


FIGURE 1. A pictorial representation of the paper's flow.

while also supporting enhanced Mobile Broadband (eMBB) and massive Machine-Type Communications (mMTC) (one of the three services defined by 3rd Generation Partnership Project (3GPP) for 5G NR deployment) [30]. As defined in International Mobile Telecommunications (IMT)-2020 [30], 5G technology offers a spectrum that ranges from 400 MHz to 71 GHz. Usage scenarios include eMBB with a focus on 10x capacity and Extended Reality (XR), mMTC with 10x capacity for the IoT and uRLLC tailored to IIoT and

Critical IoT. In terms of positioning, it offers an accuracy of up to 30 cm, with an accuracy of less than 10 cm in 5G Advanced (Release 18, 19, 20) [31]. In terms of NTN VSAT and UE, 5G utilizes S band (2-4 GHz) for uplink, L band (1-2 GHz) for downlink. 5G innovations include NR V2X, MEC in 5G Core, Native Edge Apps and the integration of Artificial Intelligence (AI)/Machine Learning (ML) such as RAN Intelligent Controller and Network Data Analytics Function. Its infrastructure includes virtualized

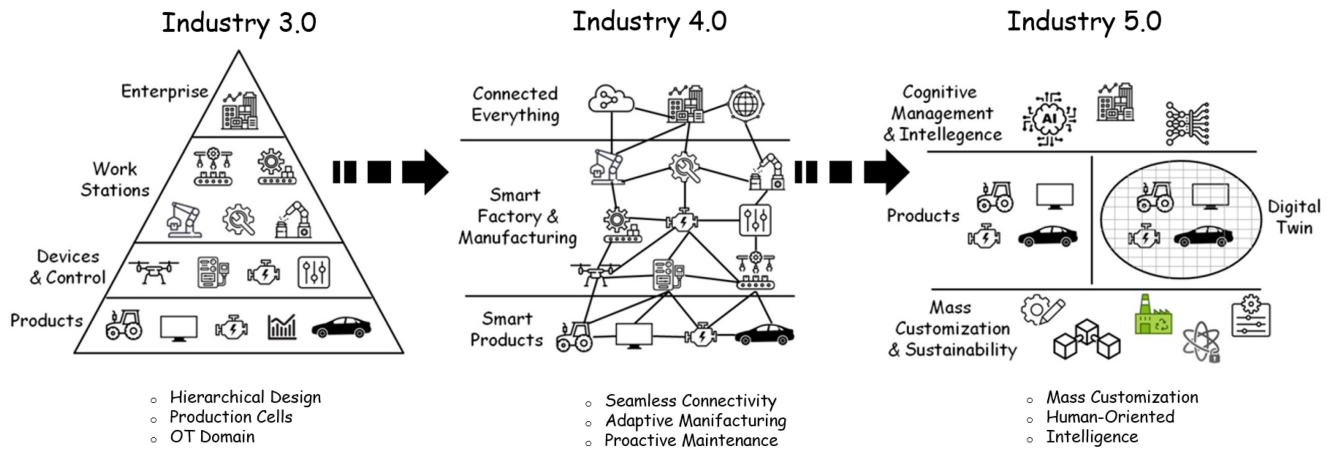


FIGURE 2. Evolution from the old industry 3.0 to new industry 4.0 and future industry 5.0.

Cloud-native Network Functions (CNFs), a disaggregated architecture with virtualized Radio Access Network (vRAN) and the implementation of end-to-end network slicing on 5G Core. In terms of security, a new security foundation is created that focuses on IoT security and promotes best practices, especially in DevSecOps [32].

In many ways, 5G has played a role on the wireless side that is similar to what Time Sensitive Network (TSN) accomplishes for wired connectivity. It served as a universal, globally standardized technology that brings together different types of services and enables the expansion of wireless connectivity to other areas of factory floor communication. However, TSN also has an additional role to play in the context of 5G [33]. Industrial networks are usually long-term installations where the majority of factories already deployed and operating. Introducing new technologies into existing brownfield installations can be a slow and challenging process. TSN is expected to trigger a redesign of building practices, with the aim of incorporating it into industrial brownfield networks whenever feasible [34]. By linking 5G with TSN as its wireless counterpart, TSN presents a market opportunity to facilitate the transformation of brownfield networks [35]. This alignment between the 5G architecture and TSN underscores their compatibility and the potential for mutual reinforcement.

3) THE ROLE OF 6G IN ADVANCING INDUSTRIAL AUTOMATION CAPABILITIES

In the realm of industrial automation, the advent of 6G technologies brings forth a myriad of key capabilities that are pivotal for transforming the landscape. As defined by the International Telecommunication Union (ITU-R) and in the context of IMT 2030, the key capabilities of 6G set to redefine communication technology standards [36]. The peak data rate of 1000 Gbps in the downlink and a user data rate of 1000 Mbps represent a colossal leap in data transmission capability [37]. This unprecedented speed is complemented by ultra-low latency of just 0.1 ms, enabling near-instantaneous communication essential for real-time

industrial applications. Spectrum efficiency, an important metric, is expected to be 10 times higher than 5G, measured in bits per second per hertz. This improvement enables more efficient use of the available frequency spectrum and takes into account the diverse communication requirements of industrial devices [38]. Designed to support mobility speeds up to 1000 km/h [39], 6G is well-suited for high-speed applications within smart factories or logistics environments, facilitating rapid and reliable automated vehicle operations.

Furthermore, the connection density of 6G is a remarkable 10 million devices per square kilometer, reflecting the ability to accommodate a large number of interconnected devices in an industrial environment. This high density is particularly important for the proliferation of the IoT and the integration of a variety of sensors and devices necessary for advanced manufacturing processes [40]. Additionally, the energy efficiency of 6G, which is 300 times more efficient than its predecessor [41], [42], aligns with the increasing importance of sustainable and energy-efficient communication solutions in the industry. The impressive capacity of 6G to handle 1000 Mbps per cubic meter of area traffic illustrates its ability to manage extensive data loads in densely populated industrial environments [43]. Overall, the capabilities of 6G set new benchmark for wireless communication and have the potential to enable critical industrial information functions such as ultra-low latency feedback, large-scale sensor integration and predictive analytics, which are crucial for the development of digital production environments.

B. FROM INDUSTRY 4.0 TO INDUSTRY 5.0

1) EVOLUTION OF INDUSTRY 4.0

The advancement of digitalization in the manufacturing system is anticipated to bring about increased flexibility and efficiency, transforming manufacturing into a cyber-physical production system [44], [45]. Figure 2 shows the evolution from the old Industry 3.0 (which is hierarchical) to the new Industry 4.0 (which is fully connected) and the future Industry 5.0 (which is built with mass customization and intelligence). The transition from old industry to new

industry and future industry is characterized by various technological advances, changes in production processes, effects on the workforce, challenges and the potential for future progress. Industry 4.0 also introduces adaptive manufacturing that enables rapid changes in production to accommodate variations in product design or customer requirements and enable the production of highly customized and personalized products. To realise this vision, it is essential to create seamless connectivity throughout the factory and eliminate isolated islands of connectivity at the factory floor level, as shown in the center of the Figure 2.

Industry 4.0 introduces the concept of smart factories, in which technologies such as the IoT, AI and data analysis are seamlessly integrated. This integration allows for real-time monitoring and control of manufacturing processes. Industry 4.0 relies on the IoT to connect a variety of physical devices and sensors built into machines, equipment and products. These devices can communicate with each other and create a network of interconnected *things* using various technologies such as BLE, Zigbee, WirelessHART, Institute of Electrical and Electronics Engineers (IEEE) Std 802.15.4 TSCH, 6TiSCH, Low Power Wide Area Network (LPWAN) (LoRa, Sigfox, NB-IoT, LTE-M) and RAW [46]. IoT sensors and devices are deployed across the production floor to collect real-time data from the industrial environment [47]. These sensors monitor various aspects of manufacturing, including machine performance, energy consumption, and product quality. This data contains information about the performance of machines, production processes and the condition of devices. The huge amounts of data generated by IoT devices are processed and analyzed using AI algorithms [48]. AI enables machines and systems to learn from this data, recognize patterns and make intelligent decisions without human intervention. These technologies can predict when equipment is likely to fail based on historical data, enabling proactive maintenance [49]. This approach reduces unplanned downtime, increases the reliability of the machines and extends the service life of the devices [50].

2) THE INDUSTRY 5.0

In the context of Industry 4.0, activities such as production line monitoring, including tasks such as collecting temperature and humidity data from sensors, can reasonably tolerate relatively higher latency times, often in the range of 50 to 100 ms. However, when it comes to real-time control of machines and robots, the performance requirements become even more stringent. These tasks require extremely low latency, such as $100\mu\text{s}$ [51], [52]. Industry 5.0 is still under development and in its early stages, but it aims to build on the foundations of Industry 4.0 and introduce new technologies that can further transform the industry [53]. Some of the expected key technologies are as follows: Quantum computing represents a paradigm shift in computation. Therefore, quantum computing has the potential to revolutionize data analysis, optimization and simulation processes in industry. It can significantly accelerate problem

solving and decision making, especially in complex scenarios of industrial automation. In Industry 5.0, blockchain is intended to improve trust and transparency in supply chains, transactions and data exchange. Smart contracts enabled by blockchain will automate and enforce agreements, reducing the need for intermediaries and streamlining processes. XR in Industry 5.0 can revolutionize maintenance, training and collaboration. Workers can use AR to access real-time information, instructions and visualizations while performing tasks. This technology improves situational awareness, reduces errors and improves efficiency. Improved connectivity through B5G/6G will enable faster and more reliable data exchange, enabling real-time monitoring, control and collaboration in Industry 5.0. This is critical for applications such as autonomous systems, remote control and connected industrial ecosystems. Finally, advanced IoT devices with enhanced sensing capabilities and embedded intelligence will help create a more responsive and adaptive industrial environment. These devices will enable data-driven insights, predictive maintenance and improved overall efficiency.

3) KEY DIFFERENCES BETWEEN INDUSTRY 4.0 AND INDUSTRY 5.0

The key differences indicated in the Industry 5.0 paradigm when compared with the Industry 4.0 are important to be identified in the context of mobile communication for industrial automation [54]. The human-oriented approach is one of the key differences between Industry 5.0 and Industry 4.0. The extension towards human-oriented industrial processes can be identified in different application contexts of Industry 5.0. In addition, the envisioned objectives of Industry 5.0 include driving the manufacturing processes towards global sustainability [55]. Mobile communication is required to align with the objectives of Industry 5.0. Industry 5.0 will facilitate more sustainable, resilient, and collaborative industrial processes than Industry 4.0. Figure 3 reflects the key differences between Industry 4.0 and Industry 5.0 illustrating Industry 5.0's focus on advancements like sustainability, human-robot collaboration, and quantum computing alongside established concepts such as big data, automation, IoT, AI-assisted manufacturing, and production monitoring. Quantum computing promises transformative advances for Industry 5.0 by optimizing supply chains, developing advanced materials, improving machine learning for predictive maintenance, enhancing cybersecurity through quantum encryption, and enabling efficient real-time data analysis, revolutionizing various aspects of industrial automation.

C. B5G DEPLOYMENT OPTIONS

The deployment of a B5G system can take different forms. If an industrial user can acquire local access to dedicated spectrum, they can deploy a standalone local B5G system. This standalone 5G network has the potential to interwork with a public network, enabling functionalities such as roaming. Another approach is to utilize federated network

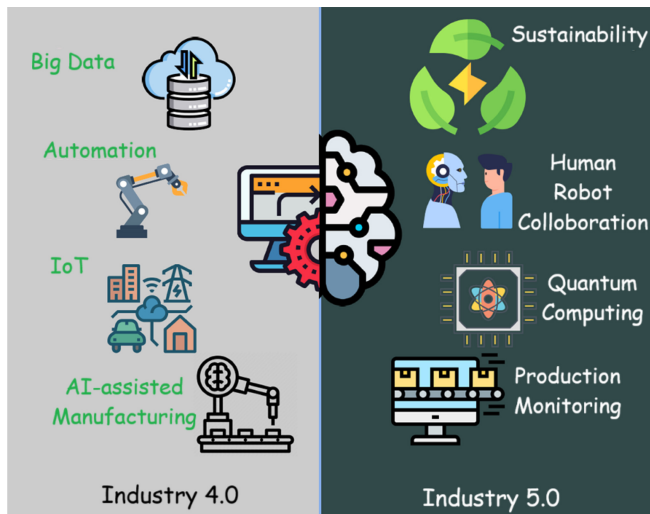


FIGURE 3. Key differences between Industry 4.0 and Industry 5.0.

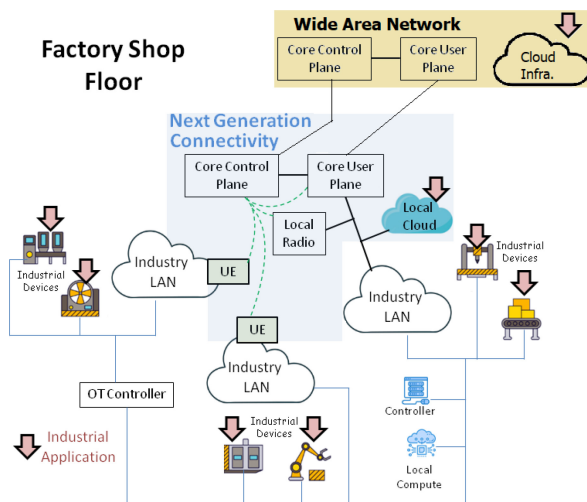


FIGURE 4. 5G non-public network interworking with public wide-area network.

slicing, where a logical network slice is created based on the physical infrastructure of two or more networks. This allows for enhanced flexibility and customization in meeting the specific requirements of industrial applications. A local B5G system can also be implemented as a non-public network service, which is offered by a public Mobile Network Operator (MNO) at the industrial location, as illustrated in Figure 4. In this setup, certain parts of the network infrastructure are typically deployed onsite. By enabling onsite data breakout, low latency can be achieved, and data privacy policies can be enforced to ensure that information does not leave the site. The core control functionality can be provided either from the MNO's external sites or partially or fully onsite to support local survivability and operational requirements [56]. While critical communication services are maintained onsite through local breakout, some other functions may utilize external termination of data sessions.

An alternative approach is to combine a standalone local network with a public MNO network to create a non-public network service that spans both network domains. In this scenario, an industrial user can deploy a local network onsite, which, when integrated with the public network infrastructure, enables the provision of a non-public network service through federated network slicing. This combination allows the local deployment to enhance the resilience and performance of the public network by providing localized coverage, increased availability, higher capacity, and additional computing resources [57]. The integration of the two networks strengthens the overall connectivity and capabilities available to the industrial user. A local network can also offer neutral-host capabilities by extending a public network onsite in addition to providing a standalone local network. To achieve this, network sharing solutions such as multi-operator core network or multi-operator radio access network can be implemented. In shared network approaches, a resource management solution is essential to ensure guaranteed resources and performance for the various supported networks or network slices. Both local and public network providers can benefit from a network sharing solution [58]. The local provider can offer a free local site to the MNO, while the MNO can provide its spectrum resources for the network. Since the same base stations can support both public and private services, improved coexistence between the local and public networks becomes feasible. Furthermore, a shared solution can be motivated by different service requirements. For example, a public MNO may deliver conventional enterprise services on the industrial site, such as telephony, mobile broadband, and IT connectivity, while the private standalone local network is dedicated to local industrial OT connectivity.

D. POSITIONING AND TIMING

By determining the location of mobile devices like Intelligent Transportation Systems (ITS), smartphones, tablets, and other production resources that are required in the factory, positioning or localization in industrial environments enables modern industrial applications to support the evolution of smart factories. Dynamic seamless multi-connectivity must be fully integrated for proper positioning computation [59]. Location determination can be carried out using Radio Frequency (RF), camera or Light Fidelity based positioning techniques and parameters such as measuring the arrival time of the received signal, transmission time, arrival angle, etc. The deployment of the nodes and their distribution within the factory floor area have a significant influence on the positioning accuracy, which is a crucial factor in the conclusion of any industrial positioning technique. This dependency means that there is always a trade-off between the positioning deployment's complexity with cost and the position estimation's potential accuracy.

On the other hand, randomly deployed IIoT sensor nodes produce distinct local clocks as a result of time offset and clock drift. For the time synchronization aspects, two critical

TABLE 3. Summary on benefits, key challenges and possible solutions for industrial automation architecture with mobile networks.

Research	Benefits	Key Challenges	Possible Solutions
OT Systems & 5G/B5G/6G	Design a mobile network that supports a wide range of use cases with stringent requirements. [1], [6], [28]	Industrial deployments where a fixed communication is not feasible due to physical constraints. [11], [56], [61], [62]	Deliver connectivity for the 24x7 production line with reliability and latency. [44], [50], [63], [64]
	Provide a platform for high volume, high reliability and always on manufacturing/production. [7], [15], [30], [40]	Hierarchically managed subsystems to ensure reliable and efficient communication. [26], [28]	Constitute a flexible architecture for enterprise grade industrial connectivity. [18], [38]
From Industry 4.0 to Industry 5.0	Efficient and secure life-cycle management readiness for 5G. [65]–[67]	Advances are required to obtain a 6G compatible IIoT ecosystem. [36], [37], [41], [43]	Adopt hybrid architectures with human-centric technologies of Industry 4.0 automation and Industry 5.0 personalization. [3], [57]
B5G Deployment Options	Guarantee traffic flows in a guaranteed time window within and between industrial device. [16], [19], [68], [69]	Enhance network energy consumption performance and reduce interference. [12], [25], [29], [47]	Introduce time sensitive features for communication latency with high availability. [33]–[35], [70], [71]
Positioning & Timing	Provide standard times and synchronization mechanisms for deterministic communication. [60]	Determine the accurate positioning and localization of devices across factory. [59]	Efficient management of devices, digitized design, factory production, and automation processes. [45], [72], [73]

difficulties must be taken into consideration and overcome in order to satisfy the stringent requirements of time-sensitive industrial applications. An effective symbol timing synchronization technique is first required in order to enable precise timestamp message recognition and decryption at slave nodes. Another issue is that, in order to increase accuracy, the time synchronization approach needs to account for clock drift during clock-offset measurement [60]. A summary for the related work and research activities of the industrial automation architecture section is given in Table 3.

III. INDUSTRIAL APPLICATIONS AND USE CASES

In this section, we look at various industrial applications that use modern communication technologies. From real-time wireless sensing to the implementation of Generative AI (GenAI), each subsection explores the transformative impact of communication advancements on specific aspects of industrial processes. These applications centred on industrial information processing and coordination - such as motion control, digital twins and cloud robotics - demonstrate the targeted integration of advanced communication technologies that support latency-sensitive and adaptive industrial operations.

A. WIRELESS USE CASES

Wireless real-time sensing has diverse potential applications in industrial automation. It can gather and store raw machine, Enterprise Resource Planning (ERP), IIoT, and controller data from end-systems which could help in extracting information and creating knowledge. The resulting knowledge can be utilized to take useful actions at the manufacturing floor, such as improving production efficiency by redesigning processes based on customer feedback or increasing resource supply for customized flows from the production warehouse in zero-defect manufacturing [74].

In the context of factories of the future, motion control emerges as a significant use case, as outlined in Chapter 5.3.2 of [75]. Motion control is an essential component of automation applications and is essential to many industrial operations, such as printing machines and industrial robotics. Motion control basically entails synchronizing the movements of several actuators to accomplish particular functions. Electronic motors, which can be categorized in a number of ways, such as AC/DC, stepper/servo/hybrid stepper, etc. are the examples of common types of actuators in industries. All classes of motion control follow the same principles, notwithstanding the diversity of motor types. When implementing motion control applications that demand a high degree of precision and accuracy, closed-loop control methods are usually used [76]. Motion control systems can be logically divided into several components: (i) The physical actuator, also known as the motor, which is responsible for generating the desired motion. It is typically equipped with one or more sensors, such as encoders, to provide feedback on parameters like speed and position. (ii) The driver, also referred to as an inverter, serves to control and regulate the power supplied to the actuator. (iii) The motion controller is responsible for executing high-level motion commands and coordinating the operation of the actuator and driver. (iv) The Programmable Logic Controller (PLC) functions as the central control unit for the overall system, coordinating various processes and components.

The functional breakdown of the motion control system can be visualized in Figure 5. The motion control architecture comprises several communication patterns, which are described below and correspond to the numbered components in Figure 5:

- The PLC communicates higher-level commands, such as *Start*, *Stop*, *Do xxx turns* or *Go to position xy*, to the motion controller. This communication typically imposes less stringent requirements.

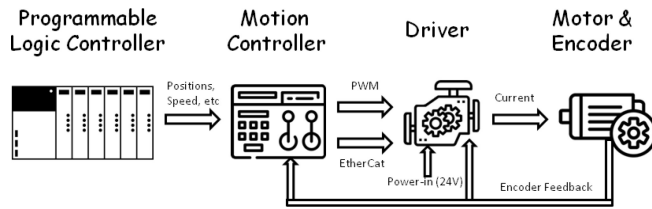


FIGURE 5. Logical function split in motion control applications.

- The motion controller periodically generates set points, which could be values like speed or torque, for the driver. This can be achieved through different methods, including: (a) Pulse-width modulation, which is not a communication technology but a control technique. (b) Communication protocols like EtherCat or Profinet Isochronous Real-Time (IRT) or similar, which support very low cycle times, often below 1 ms.
- The driver receives the set points from the motion controller and feeds currents to the motor, providing the necessary energy supply. This process does not involve communication technology but is part of the control loop.
- The encoder (sensor) provides feedback to either the driver or the motion controller, depending on the motor and encoder type. The feedback can be analog or transmitted using communication technologies such as EtherCat or Profinet IRT, with similar requirements for cyclic closed-loop set point transmission and feedback.

In the context of [75], motion control applications often involve a single motion controller controlling multiple actuators, such as multiple motors within the same machine. The requirements for these motion control applications, specifically addressing the closed-loop system consisting of the motion controller, driver, and encoder, are outlined in Table 4. According to the [75], it is emphasized that the occurrence of two consecutive packet losses is deemed unacceptable, highlighting the need for reliable communication. Additionally, a high level of synchronicity with a jitter below 1 microsecond is essential among all devices involved in the motion control system. This tight synchronicity enables the sampling of distributed encoders and the application of new set points from the motion controller to drivers at synchronized sampling points. This type of synchronized communication is known as isochronous communication, wherein the motion control program, actuators, and encoders are synchronized with the communication cycle timings dictated by the specific communication technology, such as Profinet IRT. Such synchronization ensures minimal and predictable latencies by utilizing timed channel access.

Motion control equipment vendors often integrate multiple functionalities into a single physical entity. An example is illustrated in Figure 6, showcasing the offering from an example motion control manufacturer. At the upper *Control Level*, they employ a consolidated PLC and motion controller (Logic & Motion) alongside a human-machine interface.

TABLE 4. Motion control requirements as defined in TR 22.804 of 3GPP [77].

Application	# of Sensors \ Actuators	Message Size	Cycle Time	Service Area
Printing Machine	> 100	20 byte	<2 ms	100m x 100m x 30m
Machine Tool	~ 20	50 byte	<0.5ms	15m x 15m x 3m
Packaging Machine	~ 50	40 byte	< 1ms	10m x 5m x 3m
Automated Guided Vehicle	~ 4	256 byte	40ms to 500ms	10m x 10m x 10m
Tool Localisation	~ 4	200 byte	60sec	10m x 10m x 10m
VR Tablets	~ 8	<256 byte	17ms	10m x 10m x 10m

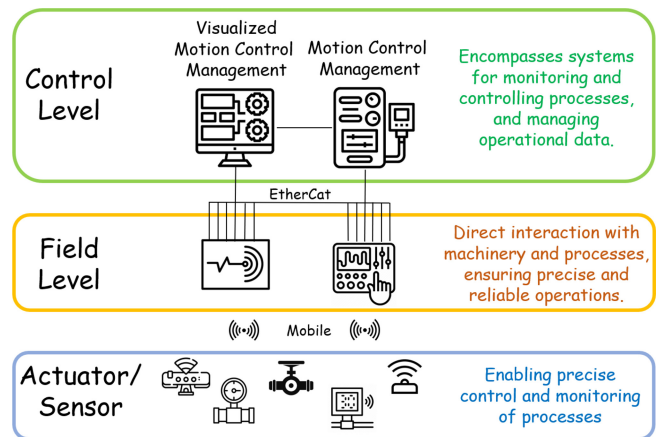


FIGURE 6. An example of structure of different product manufacturing levels in industry automation architecture.

This controller receives input from IO devices (3) and transmits set points to the servo inverter (2) using the EtherCat protocol at the *Field Level*. Another new trend is the integration of encoders, drivers and/or motion controllers directly into the motor itself [78]. In certain scenarios, multiple motion controllers may be employed to control different subsets of drivers within the same application [79]. Achieving coordinated motor movement in such cases necessitates communication between these separate motion controllers. In [75], this type of communication is referred to as “Controller-to-Controller (C2C)”, with assumed cycle times ranging from 4 to 10 ms [75]. Similarly, stringent requirements for synchronization apply at the C2C level, demanding a jitter below 1 microsecond and allowing payload sizes of up to 1 kB.

To ensure safety in motion control systems, an additional functional safety control is typically implemented. This safety control operates alongside the primary closed-loop system used for motion control. It can be achieved through the integration of safety functions within the motion

control components or the use of dedicated hardware. Communication protocols like ProfiSafe are employed for this purpose. One common safety requirement is the implementation of Safe Torque Off (STO) as defined by the IEC 61800 standard. STO dictates that in the event of any detected error or safety issue by the PLC or an additional safety PLC, power delivery to the motor must be immediately halted. For instance, an STO can be triggered by pressing an emergency button. 3GPP study in [75], explains that for functional safety, a strictly cyclic data communication service is necessary between the relevant components. If connectivity is disrupted, an emergency stop is activated, even if no actual safety event has occurred. Different use cases may have varying requirements, such as cycle times ranging from 4 ms to 12 ms, packet sizes of 40 to 250 bytes, and tolerable jitter in transmission [75]. Additionally, safety functions may be implemented in different components of the motion control architecture. In conclusion, motion control systems involve four distinct types of communication: (i) The lowest level communication is the closed-loop interaction between the motion controller, driver, and encoder. (ii) Controller-to-controller communication is required when multiple motion controllers are used to coordinate motor movements. (iii) Functional safety communication ensures the implementation of safety measures alongside the primary motion control system. Protocols like ProfiSafe are utilized for this purpose. (iv) Communication between the PLC and motion controller facilitates the exchange of higher-level commands and set points for motion control operations.

B. DIGITAL TWIN CONCEPT

A digital twin (usually an AI model) mimics and continually monitors the behavior, functionality, and operational state of a real-world system or object. The concept of the *digital twin* was initially developed for use in manufacturing and industrial settings, to simulate and enhance intricate systems like machinery and production procedures. For example, a machine's digital twin can be utilized to optimize production schedules, reduce downtime, and predict when the unit will need maintenance. A digital twin is a comprehensive and up-to-date representation of a physical asset, process, or system that can be used to optimize its design, improve its functionality, and increase its efficiency [80].

While traditional digital twins (DTs) provide a static or semi-dynamic mirror of an asset, a cognitive digital twin (CDT) goes a step further by embedding on-the-fly learning and reasoning capabilities [81]. In other words, a cognitive twin continuously adapts its internal models using streaming sensor data, real-time simulation, and AI inference at the edge or cloud. By combining high-fidelity physics-based simulations with machine learning algorithms, a cognitive digital twin can

- Self-learn from newly arriving data (e.g., automatically updating its model parameters when the machine's mechanical characteristics drift over time).

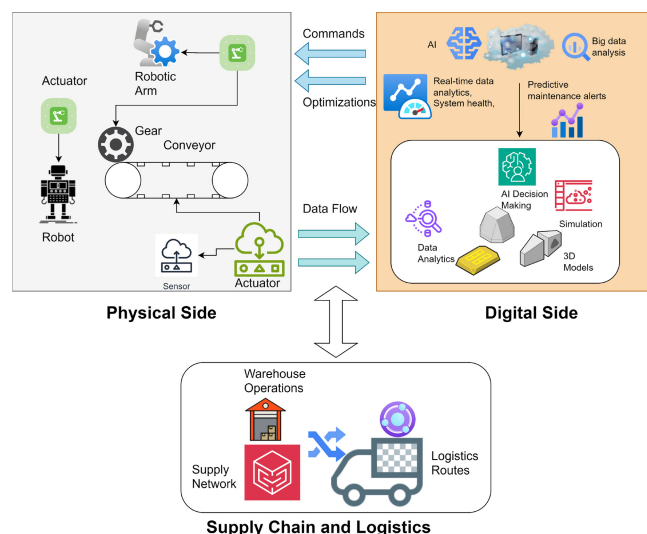


FIGURE 7. Digital twin integration in industrial automation.

- Predict future states (e.g., forecasting remaining useful life or identifying emerging failure modes) without a separate “offline” retraining step.
- Recommend corrective actions (e.g., adjusting operating setpoints or scheduling maintenance in advance of a breakdown).

In practice, this means that the digital replica is not merely descriptive but becomes an autonomous agent capable of reasoning, “what-if” experimentation, and synthesizing new training data on the fly. Henceforth, whenever we refer to “digital twin” in this manuscript, we implicitly include both static twins and their AI-driven extension-cognitive digital twins-which leverage edge/6G connectivity for real-time feedback and closed-loop control.

The concept of digital twin has gained significant popularity in industrial settings. Figure 7 shows the integration of the digital twin for industrial automation. It involves creating a digital representation of a physical asset or an entire factory, capturing data from the asset or process, and using analytics to analyze its past and current behavior [82]. Additionally, digital twin can predict and prescribe future behavior based on the gathered data. By conducting various simulations on the same product, computational efficiency can be improved by sharing relevant data between simulations instead of manually generating separate and incompatible inputs for each case [83]. While there are ongoing research inquiries related to the digital twin concept, the specific answers may vary depending on the asset being modeled. A digital twin in industrial cloud robotics can provide important information and insights about the physical unit, including the ability to detect potential problems and inefficiencies, forecast performance and aid in decision-making [84]. By providing real-time data and insights about physical assets, the digital twin helps industrial automation systems improve decision-making, speed up workflows and increase overall operational efficiency [85]. When utilizing digital twin in simulators,

one objective is to receive feedback information from the physical twin, such as haptic or force sensor data, and apply it to the digital twin [86]. To maintain synchronization and enable real-time interaction between the twins, it is crucial to have low communication latency. Typically, a millisecond-level latency is required, with a maximum tolerance of around 5ms. Achieving such low and deterministic delays can be facilitated by utilizing 5G TSN [71] and uRLLC solutions [87]. These technologies provide the necessary capabilities to ensure efficient and reliable communication between the digital and physical twins, enabling seamless interaction and synchronization in real-time [70].

In this zero-delay environment, a cognitive digital twin leverages ultralow-latency 6G (or B5G) at the edge to run inference engines that continuously refine its internal state. For example, by ingesting vibration and temperature streams in real time, the cognitive twin can update its prognostic model in-situ and immediately forecast the machine's remaining useful life rather than relying on offline data dumps. Underlying ML/AI modules (e.g., deep autoencoders for anomaly detection or recurrent neural networks for remaining-life prediction) execute at the network edge, co-located with 6G micro-data centers. This edge-level reasoning ensures that any deviation from normal operating patterns triggers near-instant alerts to the physical device or control system, closing the loop within sub-millisecond timescales. In doing so, cognitive digital twins become true real-time decision agents rather than passive observers.

DTs and CDTs enable advanced industrial automation by leveraging IoT interoperability, edge-cloud convergence, and 6G technologies. DTs/CDTs integrate heterogeneous IoT data from protocols like BLE, Zigbee, and LPWAN (e.g., MQTT, CoAP) to ensure real-time updates for predictive maintenance and quality control in Industry 4.0/5.0. 6G's ultra-low latency (0.1 ms), high data rates (1000 Gbps), and 10M devices/km² density (p. 7-8) facilitate seamless DT/CDT synchronization and secure DTN orchestration, supported by edge computing for time-critical tasks (e.g., anomaly detection) and cloud platforms for large-scale analytics (e.g., process optimization). SOTA solutions like NFV and SDN (p. 3) enable IoT virtualization and orchestration, supporting dynamic resource allocation for DT-driven zero-defect manufacturing. Case studies include DTNs in automotive manufacturing for real-time defect detection using 6G-enabled IoT orchestration and CDTs in logistics for predictive supply chain optimization via edge-cloud integration, enhancing scalability and efficiency in smart factories.

To anticipate potential future scenarios, it is essential to model the key physical assets and control functions within the digital twin. However, for complex systems with multiple objects and numerous degrees of freedom, it becomes difficult to look too far into the future. Usually, only the next few seconds can be reliably predicted, especially in dynamic environments where external influences can have significant effects [88], [89]. In controlled environments such as factory

floors, longer-term estimations in terms of minutes might be feasible. In the context of human-robot collaboration, the predicted future can be exceptionally short, given the need for real-time responsiveness and close interaction between humans and robots. To set up real-time digital twin, cloud-based solutions, especially edge cloud solutions, are usually used, especially when AI-based controls are integrated [90]. It is critical to achieve all of the above goals with a user-friendly system that increases reliability, availability, risk reduction, maintenance cost savings, and production improvements [91]. In addition, the integration of AR and VR with the concept of the digital twin has the potential to significantly impact network management in the merged IT and OT domains [92]. With AR and VR technologies, devices can be managed and adjusted remotely, giving users the feeling that they are physically in the same room. In addition, technical documentation, instructions on how to use equipment, assistance with repairs and troubleshooting can be made available to on-site personnel via smart glasses, tablets and other devices [93].

In industrial practice, cognitive digital twins already demonstrate tangible benefits:

- *Steel Manufacturing:* A leading pipe mill built a cognitive twin that ingests streaming sensor data (pressure, temperature, spindle current) and continuously refines a predictive model. This twin can detect early micro-fractures, prevent catastrophic line stoppages, and optimize rolling schedules all without manual model retraining [94].
- *Automotive Assembly:* A cognitive twin hosted on an edge-cloud platform combines LiDAR and robotic arm telemetry to predict joint wear in real time. It dynamically tunes the robot's force-control setpoints, extending tool life by up to 20% while reducing scrap rates by 15%.
- *Pharmaceutical Batch Processing:* By applying cognitive twin technology to clean-in-place (CIP) systems, a global pharma plant learned to optimize cleaning cycles, reducing water and chemical usage by 25% while ensuring regulatory compliance [95].

These examples show how AI-enabled digital twins can support predictive maintenance and adaptive control in manufacturing environments functions that benefit from the ultra-low latency and scalability of next-generation wireless networks.

C. CLOUD ROBOTICS

Cloud robotics is a branch of robotics that uses cloud computing to improve robot capabilities and performance [96], [97]. It is a major focus in industrial robotics research and robotics in general [98] and explores the use of various cloud technologies to extend the tasks of robotics and improve the flexibility and capabilities of the overall system. Robots in cloud robotics are linked to a remote cloud-based network, giving them access to

TABLE 5. Summary on benefits, key challenges and possible solutions for industrial applications and use cases with mobile networks.

Applications	Benefits	Key Challenges	Possible Solutions
Wireless Use Cases	Enhanced asset management through wireless interface for higher-priority measurements. [75], [89]	End-to-end tracking and monitoring goods, production and asset data. [74]	Deploy private mobile networks enables low-latency, reliable, and secure wireless communication. [86], [101]
Digital Twin	Develop augmented/virtual reality for applications, such simulation of factory processes. [10], [92], [93]	Provides cloud solution development and integrates IIoT to streamline business processes. [102], [103]	Leverage ultra-reliable low-latency communication for real-time synchronization of systems with digital twins. [93], [98]
Cloud Robotics	Integrate cloud computing technology with robotics and industrial automation systems. [97], [104]–[107]	Detect issues and respond with computing in cloud to improve performance and lower costs. [108], [109]	Utilize edge computing for cloud robotics by enabling low-latency processing and real-time coordination. [108], [109]

massive amounts of data, computational resources, and software applications. Therefore, cloud-fog-edge computing is a crucial technological advancement for cloud robotics which involves utilizing cloud computing resources to support robotic systems. Using a combination of real-time data and machine learning algorithms, the cloud network enables robots to perform difficult jobs and make intelligent decisions. Moreover, by accessing vast amounts of data and computing power through cloud robotics, robots can perform more intricate tasks and make better-informed decisions [99]. Therefore, cloud robotics can facilitate human-robot collaboration, enhance manufacturing processes, and increase the efficiency and productivity of industrial systems.

Flexibility is a crucial requirement in the context of Industry 4.0, as it enables cost-effective and customized production by facilitating rapid reconfiguration of production lines and easy application development. However, typical industrial applications demand time-sensitive and highly reliable end-to-end communication. To meet these requirements, technologies such as 5G uRLLC [100], and edge cloud are essential. While not all cloud robotics applications necessitate real-time communication, there are several that heavily rely on it, particularly when the cloud's processing directly influences the immediate motion of the robot. Additionally, a summary of the section is given in Table 5.

IV. TECHNOLOGY ENABLERS

A. END-TO-END RESOURCE ASSIGNMENT

In industrial automation and other ultra-low latency applications, understanding the specific communication and control requirements in each context is crucial. Various methods are used for end-to-end resource allocation [110], each suited to different applications:

- *Dedicated vs. Shared Resources:* In industrial automation, such as motion control and real-time sensing, dedicated communication resources are crucial for ensuring ultra-low latency and high reliability. Deterministic communication via EtherCAT or Profinet IRT enables sub-millisecond cycle times essential for motion control [111]. Conversely, applications like VR or autonomous driving might use shared resources if they still meet the latency and reliability needs. Network slicing in 5G networks, for instance, can create virtual

lanes that ensure required performance more cost-effectively.

- *Synchronization Requirements:* Industrial applications often necessitate extremely tight synchronization between devices, such as motion control systems where jitter must be less than 1 microsecond, ensuring simultaneous sampling and actuation [112]. Applications like emergency services communications require ultra-low latency but can tolerate slightly higher jitter and less stringent synchronization, focusing more on speed than precise timing.
- *Communication Reliability and Safety:* Safety is paramount in industrial environments. Protocols like ProfiSafe are designed for safety-related communication to ensure high integrity and availability for vital and machine-critical functions. Autonomous guided vehicles, for instance, need reliable communication for safety but balance immediate response needs with centralized decision-making latency.
- *Real-Time Data Handling:* Real-time control systems in industrial automation must process data with minimal delay, supporting real-time data acquisition, processing, and actuation. This often integrates with cloud and edge computing to manage data flow efficiently. Conversely, ultra-low latency applications like streaming services may use predictive algorithms to smooth out intermittent network delays, maintaining user experience despite potential fluctuations.
- *Network Architecture/Slicing:* Network architectures in industrial settings are typically hierarchical and segmented to enhance security and manageability, supporting localized control loops with ultra-low latency while connecting higher-level functions to broader network services, including cloud analytics or remote monitoring [113]. Other applications might use decentralized or mesh networks to distribute data and processing demands more broadly, enhancing resilience and scalability.

B. MACHINE LEARNING

Within the broader field of AI, ML is the application of statistical techniques to enable computer systems to “learn” from data without explicit programming. To boost

automation and efficiency, for example, ML can be used to automate the creation of analytical models and offer techniques for designing data-driven reasoning [114]. This can be applied to many different elements of a system. Additionally, ML may alter network consumption patterns, defining new communication requirements for industrial applications. For instance, the underlying communication layer may be required to transfer the collected data samples before the next sample enters the buffer if crucial monitoring data must be gathered and provided to the control element every say T ms [115]. A digital energy appraisal enabled by smart sensors (that can be mounted on electric motors pumps and general machinery) can help to perform data analytics (using power consumption, speed vibration frequencies and temperatures) which can rectify potential faults well in advance of possible breakdowns or help energy efficiency.¹ A cement factory condition monitoring system can perform preventive maintenance and improve the performance and efficiency of the smoke fans by monitoring operational variables and health indicators (e.g., temperature, vibration, availability, number of starts, running conditions, current, and faulty events).²

1) ANOMALY DETECTION FROM MACHINE CONDITION

Detecting machine state anomalies in an industrial automation scenario requires a combination of data acquisition, preprocessing, algorithm selection, evaluation, and real-time monitoring [116], [117], [118], which are discussed in more detail below: (i) *Collect data*: The first step is to collect sensor data from the machine to be monitored. This can be done by installing sensors on the machine that measure various parameters such as temperature, pressure, vibration, and speed. The gathered data ought to span a timeframe that accurately reflects the typical functioning of the machine. (ii) *Preprocessing*: The collected data may contain errors or outliers that may affect the accuracy of anomaly detection. It is important to preprocess the data to remove these errors and outliers. This can be done by applying filters, smoothing techniques, neural networks or other statistical methods. (iii) *Choosing the right algorithm*: There are many algorithms for anomaly detection, and the choice of algorithm depends on the nature of the data and the specific requirements of the application. Some popular algorithms for anomaly detection are one-class support vector machine [119], Autoencoders, Isolation Forest and Multi-Layer Perceptron [120]. (iv) *Evaluation of the algorithm*: Once an algorithm has been selected, it is important to evaluate its performance in detecting anomalies. To do this, train the algorithm with the preprocessed data and test it with a separate test set. Performance can be measured using metrics such as accuracy, precision, recall, and F1 score. (v) *Monitor the*

machine: Once the anomaly detection algorithm has been trained and evaluated, it can be used to monitor the machine in real-time. Any anomalies detected can be used to trigger alerts or take corrective action to prevent imminent machine failures.

2) TROUBLESHOOTING SUPPORT

The main goal of troubleshooting support with ML is to improve the efficiency of the troubleshooting process by automatically identifying the causes of test failures to reduce the time it takes human troubleshooters to find the root cause of failures. These measures can help to minimize downtime and optimize system performance. The main steps to follow are similar to those described in the anomaly detection phase, despite the differences in leveraging the ML algorithms, such as decision trees, or clustering approaches [121], [122]. In addition, the ML model used provides recommendations for troubleshooting in the last step.

3) CAVITY FILTER TUNING WITH AI

Cavity filters are a type of filter used in radio base stations and networks to remove unwanted frequencies from the signal. They work by only allowing certain frequencies to pass through a small opening, while others are rejected. This helps to improve the overall performance and reliability of the network by reducing interference and noise. To properly utilize cavity filters in wireless base stations and networks, the correct frequency range for the filter must be selected, a filter with a wide passband and a high rejection ratio must be selected, and the filter must be regularly checked and maintained to ensure that it is working properly. However, physical deviations may still occur during the manufacturing process due to cross-coupling and the filters must be fine-tuned manually to fit to specifications. To mitigate this costly and time-consuming manual tuning process, previous work has investigated AI-based methods such as reinforcement learning [123], [124].

C. GENERATIVE AI

GenAI is a rapidly evolving field that has the potential to revolutionize industrial automation. GenAI offers a range of promising applications that can transform the way we design, operate and maintain industrial systems. One of the most compelling applications of GenAI in industrial automation is in predictive maintenance [125]. By analyzing sensor data collected from industrial equipment, GenAI models can detect patterns that indicate impending failures, allowing proactive intervention before costly breakdowns occur. This predictive approach can significantly reduce downtime, improve asset utilization and increase overall system reliability [126].

GenAI also holds immense potential for process optimization. By analyzing large amounts of operational data, GenAI models can uncover hidden insights and identify inefficiencies in industrial processes. This knowledge can then be used to streamline workflows, eliminate bottlenecks

¹Online: <https://new.abb.com/service/motion/data-and-advisory-services/energy-appraisal>, Available: June 2023.

²Online: <https://new.abb.com/news/detail/80450/abbs-condition-monitoring-services-help-mokra-cement-plant-save-210k-while-increasing-operational-efficiency>, Available: January 2023.

TABLE 6. Summary on benefits, key challenges and possible solutions for technology enablers with mobile networks.

Applications	Benefits	Key Challenges	Possible Solutions
End-to-end Resource Assignment	Demand of a high degree of precision with green, accurate and closed-loop feedback. [76], [78], [79], [132]	Immersive remote operations for factory shop floor reconfiguration and layout changes. [85], [133]	Utilizing software-defined networking enables dynamic allocation of resources across the network. [88], [89]
	Simplify point of failure with mobile network solutions for the motion control. [102], [134]	Represent virtually the industrial device, process, or system to simulate the behavior. [80], [82]	Implement network slicing for end-to-end resource assignment by dedicating specific network segments. [83], [90], [135]
Machine Learning	Execute the ML pipeline in production and transmit the trained model for implementation. [114], [116], [120], [122]	Self-healing and dynamic optimization of the communication network with ML analysis. [117], [119]	Integrate mobile edge computing with machine learning enables real-time data analysis and decision-making. [126], [136]
Generative AI	Generate AI models/actions by minimizing the error between the predicted and target state. [125]	Increase the power of the digital twin by generating any number of possible object states. [128]	Leverage high-speed mobile networks to process and analyze industrial data in real time. [130]

and optimize resource allocation, resulting in higher productivity and lower costs [127]. In the area of new product development, GenAI can play a crucial role in accelerating the innovation cycle. By generating novel designs, simulating material properties and predicting product performance, GenAI models can help engineers explore a wider range of design options and identify the most promising ones [128]. This can lead to faster development cycles, shorter time-to-market and more innovative products.

The impact of GenAI on industrial automation is already being felt across various industries. Siemens, for instance, is leveraging GenAI to develop predictive maintenance systems that can anticipate equipment failures in power plants and manufacturing facilities.³ Similarly, General Electric is utilizing GenAI to optimize the performance of gas turbines, maximizing energy efficiency and reducing emissions. Together with Microsoft, Siemens also introduces a generative AI-powered co-pilot system to enhance human-machine collaboration and boost productivity.⁴ These examples illustrate the transformative potential of GenAI in driving industrial automation towards a more intelligent, efficient, and sustainable future. Additionally, GenAI can generate data to create different use cases and determine corner cases in digital twin applications [129].

As GenAI models continue to mature and their capabilities expand, we can expect to witness even more innovative applications that reshape the landscape of industrial automation. From automating routine tasks and enhancing decision-making to enabling entirely new manufacturing processes, GenAI is poised to revolutionize the way we interact with and manage industrial systems [130]. The future of industrial automation is undoubtedly intertwined with the advancements in GenAI, fueled by the recent Large Language Models and their vast application landscape [131], promising a world where machines learn to create interfaces, adapt to the application at hand, and innovate ways of

communication at an unprecedented scale. Additionally, a summary of the section is given in Table 6.

V. INDUSTRIAL DEVICES

Industrial devices encompass a wide variety from minimalist single-purpose sensors to complex units such as robotic units and industrial production lines. In the literature, the definition of what constitutes an “industrial device” is still under investigation. These devices are generally divided into two categories: Sensors that record environmental or operational data, such as temperature or humidity, and actuators that change state in response to signals, such as the brightness of a lamp or the speed of an HVAC system. More sophisticated devices combine multiple sensors and actuators but communicate via a single interface, while the most complex systems can integrate several such units, essentially forming a miniature network of communicating components. These complex devices often contain microcontrollers or small computers to manage communication protocols, data processing and storage. Alternatively, some devices may use special communication modems equipped with their own processing capabilities. This essentially turns a highly complex device into a small network in which the components can interact and communicate independently of each other.

The communication requirements of these industrial devices vary greatly depending on their intended use and the critical nature of their tasks. For example, a simple temperature sensor might have minimal communication requirements, while a robot that is remotely controlled via a cloud service needs uRLLC. Networks therefore need to accommodate a variety of devices and service types within the same infrastructure. Complex devices such as a robot equipped with both a video camera for surveillance (which requires a high bandwidth mobile broadband service) and a precision manipulator arm (which requires uRLLC) are an example of where networks need to manage multiple service types simultaneously. In order to meet the market needs of the various industrial sectors, several important research questions about these devices need to be answered. These include issues such as integrating devices with different

³ Available online: <https://www.siemens.com/global/en/products/automation/topic-areas/artificial-intelligence-in-industry.html>.

⁴ <https://news.microsoft.com/2023/10/31/siemens-and-microsoft-partner-to-drive-cross-industry-ai-adoption/>.

uRLLC requirements, managing different uRLLC and non-uRLLC data streams within a single device, and ensuring Quality of Service (QoS) within these complex systems. Ensuring internal redundancy, monitoring QoS metrics and efficiently communicating these metrics to network controllers or base stations is also critical.

Another challenge is to verify whether the strict uRLLC requirements of an application, e.g., for motion control systems that require near-perfect reliability within milliseconds, can be met in practical scenarios. This is often complicated by the limitations of simulation models and the need for extensive testing in real-world environments. One approach could be the synthetic transmission of packets to test the reliability of the system before actual data transmission begins. Advanced resource allocation strategies, possibly using risk-aware reinforcement learning, could optimize the setup for modular industrial scenarios so that resource configurations can be adapted and reused in different environments. Ultimately, an industrial device is not just a standalone entity, but an integral part of a broader system that may incorporate edge computing capabilities or participate in federated machine learning, improving both computational efficiency and data protection. The design of such devices often requires separate environments for modem operation and application processing to meet the specific requirements of different applications while maintaining consistent communication standards. This segmentation allows for customization based on the complexity of the required sensing or actuation functions.

A. 3GPP COMPLIANT INDUSTRIAL DEVICES

The 3GPP device ecosystem, which includes chipsets and devices, is key in the context of industrial applications and play a pivotal role in defining standards that shape the device ecosystem for mobile communications. Recent developments have significantly improved this ecosystem, especially with the advent of 5G technologies. The development of a robust device ecosystem supporting New Radio (NR) uRLLC and IIoT capabilities has evolved over several years to develop mainstream commercial products such as smartphone modems. However, these initial uRLLC capabilities do not quite meet the stringent requirements of industrial applications. 3GPP has indeed defined a robust set of mandatory features primarily focused on eMBB applications, as eMBB was a primary target of early 5G deployments (e.g., 3GPP Release 15 and Release 16). These features focus on high data rates, improved capacity, and spectrum efficiency. While eMBB has a clearly defined set of mandatory functions, the NR uRLLC functions are not as clearly specified in terms of mandatory features. uRLLC support in 5G NR is introduced from Release 16, but its implementation is often scenario-specific (e.g., for industrial automation, vehicular communication). Optional features such as configured grant, mini-slots and preemptive scheduling improve uRLLC, but may not be mandatory for all devices. The NR functions for the IIoT are still

being developed and vary across releases. Features such as Reduced Capability User Equipment (RedCap) in release 17 are geared towards IIoT use cases by reducing device complexity and power requirements. However, compared to eMBB, earlier versions actually lack generally binding specifications for IIoT.

New Features for NR User Equipment (UE): Significant efforts were made to classify NR-UE features that could significantly improve latency reduction and reliability [137]. This collaborative work, involving key stakeholders and industry technical teams, aims to align future device features with industry needs. 3GPP Technical Specification 38.306 outlines various optional features for NR UE that can enhance performance by reducing latency and improving reliability [138]. Notable features include:

- *Short DRX Cycle:* This function enables the UE to switch to energy-saving mode more quickly during periods of inactivity and thus reduce the latency time when returning to the active state.
- *Intra-Slot Frequency Hopping for PUSCH:* Allows the UE to switch frequencies within a single slot during uplink transmissions, increasing reliability through frequency diversity.
- *TDD Multi DL-UL Switch Per Slot:* Enables multiple downlink-uplink switch points within a single slot in time division duplexing mode, which improves adaptability to different traffic requirements and reduces latency.
- *Short Bandwidth Parts (BWP) Switching Time:* Supports fast switching between BWP, allowing the UE to quickly adapt to changing network conditions, which can reduce latency and improve reliability.
- *Pre-emption Indication for Downlink:* Allows the network to preempt ongoing downlink transmissions so that higher priority traffic can be transmitted with lower latency.

Note that above features are optional and may not be supported by all devices. However, their implementation can significantly enhance the performance of NR UEs in terms of latency and reliability.

Reduced Capability Devices: RedCap - also known as NR-Light - was introduced in 3GPP Release 17 (2022) and targets use cases that require lower complexity and reduced data rates compared to traditional 5G devices [139]. This category is ideal for applications such as wearables, industrial sensors and video surveillance where device cost, size and power consumption are critical factors. RedCap devices support peak data rates below 250 Mbps, facilitating the expansion of the 5G device ecosystem to a wider range of applications [140], [141].

Fixed Wireless Access Devices: The Fixed Wireless Access (FWA) sector has experienced significant growth, with a variety of companies contributing to the 3GPP FWA device ecosystem [142]. As of November 2024, the GSA 4G-5G FWA Forum Device Ecosystem Catalogue identifies

numerous chipset suppliers, module suppliers and device manufacturers active in this space. This expansion underlines the increasing importance of FWA for the provision of broadband services, especially in underserved areas [143].

Cellular and LPWA IoT Device Ecosystems: The integration of mobile communications and Low Power Wide Area (LPWA) technologies has made a decisive contribution to the Internet of Things (IoT). The early ecosystem mainly included NB-IoT and LTE-M technologies, with all current devices and chipsets supporting FDD configuration. These technologies are generally suitable for non-latency-critical applications, indicating a gap in the market for devices that meet uRLLC demands necessary for more responsive industrial operations. A June 2024 report shows that the top five mobile module suppliers Quectel, Fibocom, Telit Cinterion, Semtech and u-blox - have a combined market share of 72% in terms of revenue [144]. This concentration reflects the competitive landscape and the rapid development of IoT devices within the 3GPP. On the other hand, the market gap between NB-IoT/LTE-M and the more demanding URLLC requirements suggests a potential for intermediate solutions such as advanced LTE devices (beyond Cat-M). These could serve a wider range of industrial applications where slightly higher latency is acceptable.

B. DESIGN ASPECTS

Today's network devices and products are primarily optimized for mobile broadband applications and are not inherently structured to meet the stringent requirements for ultra-reliability and ultra-low latency required for mission-critical functions. Mission-critical applications require rigorous test protocols to ensure that extremely high reliability standards are met, which are often not covered by standard mobile broadband tests. To overcome these challenges in uRLLC, the industry considers implementing system redundancy. For example, the integration of dual chipset modules [145] or the use of multiple UE in a single industrial device could increase system reliability [29]. These multiple UEs could operate on separate network paths or different frequency bands to ensure continuous service even if one path fails. This method increases reliability by duplicating data packets across multiple channels and using more than one active UE.

At the same time, the development of uRLLC and IIoT technologies has recently been a focal point for UE chipset and System on Chip (SoC) vendors. Recent advancements indicate significant progress in this domain: (i) *Quectel's 5G Modules:* Quectel has introduced a range of 5G modules that support uRLLC and IIoT applications. The product range includes features such as network slicing, 5G LAN (Local Area Network), Non-Public Network (NPN) support and RedCap for IIoT scenarios. These modules are compliant with 3GPP Release 16 and Release 17 standards emphasizing ultra-reliable low latency communications and improved support for industrial IoT and tested in various papers [146], [147]. (ii) *UNISOC's Tanggula V516*

Platform: UNISOC announced the Tanggula V516 platform, which is among the industry's first to support 5G Release 16 features [148]. This platform is tailored for eMBB, uRLLC, and IIoT applications, enabling functionalities such as differential protection, high-precision machine collaboration, and industrial Local Area Networks (LANs) in IIoT environments. (iii) *3GPP Release 17 Enhancements:* 3GPP has continued to enhance uRLLC and IIoT capabilities in its Release 17 specifications. These enhancements include improved hybrid automatic repeat request acknowledgment (HARQ-ACK) feedback mechanisms, support for uRLLC operations in unlicensed bands, and better multiplexing among different traffic types. These developments aim to meet the stringent requirements of industrial automation and other IIoT applications [149]. Another important aspect of interoperability in IIoT environments is the integration of Wireless Personal Area Network (WPAN) protocols such as Bluetooth Low Energy (BLE), Zigbee, WirelessHART and 6TiSCH. These protocols are often used in legacy and modern industrial deployments due to their energy efficiency and deterministic behaviour. However, they were not originally designed for seamless integration into 5G/6G infrastructures, which leads to challenges in ensuring deterministic communication, latency guarantees and centralised orchestration. In addition, next-generation SCADA systems are increasingly reliant on real-time bidirectional communication and adaptive control loops, which are often limited by the protocol translation overhead between WPAN-based sensor networks and cellular backhaul layers. The complexity of interoperability is exacerbated by the coexistence of industrial fieldbus protocols such as Modbus, Profibus and EtherCAT with IP-based IIoT stacks. These legacy protocols follow rigid polling-based communication models and offer no inherent support for cloud-native paradigms or cross-layer QoS signalling. Bridging these heterogeneous domains requires robust gateway design, protocol-aware abstraction layers, and time-sensitive edge computing architectures. Emerging trends suggest the need for unified protocol translation layers and software-defined interfaces that can seamlessly mediate between WPAN-based clusters and cellular IIoT backbones [150], [151]. However, ensuring end-to-end synchronization, security, and scalability especially for mission-critical industrial workloads remains an open research challenge.

These advancements reflect a concerted effort by chipset and SoC vendors to develop products that meet the rigorous demands of URLLC and IIoT applications, facilitating the deployment of reliable and low-latency communication networks essential for industrial automation and other critical use cases. Additionally, a detailed classification of industrial devices is given in Table 7.

VI. STANDARDIZATION AND RESEARCH PROJECTS

Standardization efforts establish a set of uniform criteria to establish compatibility and interoperability among diverse systems, devices, and communication channels to

TABLE 7. Detailed classification and overview of industrial devices in mobile network ecosystems.

Device Classification	Benefits	Key Challenges	Potential Solutions
Sensing Devices	Simple sensors for environmental data like temperature or humidity. Cost-effective and essential for monitoring basic operational parameters [137].	Limited in functionality and often cannot handle complex data processing or decision-making tasks [138].	Integration with advanced data analytics platforms to enhance utility without hardware changes [141].
Actuating Devices	Actuators that respond to commands, vital for automated control systems like HVAC or lighting systems [137].	Challenges in achieving seamless integration and synchronization with broader system controls [139].	Implement robust communication protocols and real-time control systems to enhance responsiveness [140].
Complex Integrated Systems	Devices with multiple sensors and actuators, capable of more autonomous operations and interactions within industrial settings [142].	Complex coordination and communication needs, higher demands on processing power and data handling [143].	Use of microcontrollers and dedicated communication modems to manage interactions and process data efficiently [144].
Networked Device Ecosystems	Networks of interconnected devices forming complex systems for comprehensive monitoring and control across facilities [146].	Integration challenges, high requirements for reliable and continuous communication, and data security concerns [147].	Adoption of network slicing for differentiated service delivery and enhanced security measures to protect data flows [148].

enable seamless integration across different manufacturing systems over the communication networks. The statutory organisations and private enterprises, individually or as a consortium, significantly invested in the research investigating industrial automation and relevant communication technologies that will be further developed as standardisation frameworks. The innovations and research contributions from the academia and the industry are eventually evolved as the standards upon iterative and collaborative improvements. The research organizations play a critical role in reflecting the feasibility of standardization frameworks in collaboration with the delegates from statutory institutions that implement the standards with collaborative research funding and investments for investigations related to the standardization.

This section covers the most important standardization activities and research projects relevant to mobile communication for industrial automation. Section VI-A briefly explains the significant standardization efforts from the leading Standards Developing Organizations (SDOs) and the relevance for the Industry 5.0. Section VI-B briefly explains each leading research project in the context of industrial automation and future networks.

Table 8 summarizes the most important projects in the field of industrial automation and industrial communication. Table 9 on the other hand summarizes the important standardization efforts related to Industry 4.0 Security and in the field of industrial automation architecture and applications driven by different organizations. The technical white papers and standards provide insights for the alignment of automated industries in different avenues, including automation architecture, security specification, and process management. The dates refer to the latest updates, taking into account possible unpublished versions due to ongoing standardization efforts.

A. STANDARDIZATION

Standardization holds significance in industrial automation, ensuring smooth integration and interoperability among diverse equipment and systems. This fosters a cohesive and efficient manufacturing environment. In the domain of mobile networks, standardization is crucial for enabling compatibility among different devices and networks, facilitating widespread connectivity and communication. SDOs play a crucial role in establishing common standards in both industrial automation and mobile networks. This contribution enhances reliability, transparency, innovation, and accelerates the adoption of advanced technologies across various industries.

International Organization for Standardization (ISO) is a global entity dedicated to developing and publishing international standards, aiming to ensure consistency, interoperability, and quality across diverse industries [152]. In the context of industrial automation within Industry 5.0, ISO plays a vital role by creating standardized frameworks and specifications. These standards offer a universal language and guidelines for designing, implementing, and operating automated systems. They guarantee seamless integration, compatibility, and efficiency across various industrial processes. ISO’s significance lies in its capacity to promote innovation, improve safety, and encourage global collaboration in the dynamic landscape of Industry 5.0. The Industry IoT Consortium (IIC) [153], is an open membership organization dedicated to developing the technical foundations for IIoT systems. Officially known as The Industrial Internet Consortium, IIC houses several Working Groups (WG) focused on enhancing the interoperability, technology adaptation security and trustworthiness of IIoT systems. The Open Industry 4.0 Alliance (OI4) [154], is a global alliance with a mission to accelerate the adoption of Industry 4.0 worldwide by addressing critical implementation challenges.

TABLE 8. Contribution of ongoing research projects.

	Description	Technical Aspects				Related Technologies				
		Architecture	Service Automation	E2E Service Life-cycle Management	Security	AI/ML	SDN/NFV	NS	Edge/Cloud Computing	Other Technologies
APRIL	Focused on innovating market-oriented robot prototypes for automating the manufacturing and processing of flexible and deformable materials.	L	M	H	M	L	L	L	L	H
SHOP4CF	Investigates to balance cost-efficient automation of repetitive tasks with active human engagement in adaptability, creativity, and agility, promoting a connected and automated factory model.	H	H	H	M	L	L	M	H	H
CONVERGING	Focused on creating intelligent, adaptable production systems that integrate autonomous agents, such as collaborative robots, AGVs, and human workers, to ensure harmonious, secure, and inclusive interactions within a "social industrial environment."	H	H	H	M	L	L	M	M	H
MASTERLY	Investigate on the transformation of mass production towards more customized approaches using adaptable robotic solutions, including modular grippers, mobile and industrial robots, and AI-driven control for handling diverse parts, with a focus on inclusivity and demonstrating effectiveness through testing in various production sectors.	H	H	H	L	H	L	L	H	H
REMODEL	Focused on integrating a dual-arm robot into production lines for manufacturing products with extensive wires and cables, resolving major manipulation issues while enhancing perception and interaction functions towards proven effectiveness in four industrial cases.	H	H	H	L	H	H	L	H	H
Smart5Grid	This initiative transforms traditional power grids into advanced, automated smart grids with cutting-edge communication technologies like the Smart5Grid MEC platform and 5G Core Service-Based Architecture, addressing device density, bandwidth, and ultra-low latency for efficient, secure operations, and rapid fault detection.	H	H	H	L	M	H	M	H	M
6G Flagship	This program aims to develop key 6G technologies, tools, and a test network, along with vertical applications, to advance digitalization and maintain a pioneering role in global 6G research.	H	M	M	H	H	H	H	H	M
ARCADIAN-IoT	This project aims to enhance security, trust, and privacy in IoT systems by providing users with greater control and transparency while envisioning autonomous management of these aspects.	H	H	H	H	H	H	L	H	M
TARGET-X	This initiative aims to accelerate digital transformation in energy, construction, automotive, and manufacturing sectors through extensive trials of 5G/6G technologies, focusing on KPIs and KVs like sustainability, security, and privacy, to develop new business models and strengthen the 5G/6G ecosystem.	L	H	H	L	M	H	M	H	M
6G-SANDBOX	This project aims to enhance the European experimentation ecosystem for 6G research by providing fully configurable trial networks across EU locations, supporting automated experimentation, and establishing an open repository for sharing 6G developments and APIs.	L	L	H	L	M	H	M	H	L

L Low Coverage **M** Medium Coverage **H** High Coverage

The Open Platform Communications (OPC) Foundation [155], is an industry consortium with the purpose of creating and maintaining standards for open connectivity in industrial automation devices and systems.

OPC defines standards for open connectivity, covering industrial automation devices, systems, process control, and data management. These standards include specifications for communication, alarms, events, historical data, and

TABLE 9. Recent important standardization efforts related to industrial automation architecture and applications.

SDO	Standard/ Specification/ Publication	Topics/Description	Update/publication date ⁵
ISO	ISO 16300-2:2019	This document is entitled as "Interoperability of capability units for manufacturing application solutions Part 2: Capability templates and software unit cataloguing". This document specifies a set of template definitions to describe the capability of a software unit of an automation solution that can be mapped to the functional requirements of a target manufacturing application.	2019
ISO	ISO 10303-1:2024	This document is entitled as "Industrial automation systems and integration — Product data representation and exchanges". It also consists the architectural principles and structure of the ISO 10303 series, which standardizes product information representation and exchange in the industrial applications including the scope, architecture, data specification methods, and various components like integrated resources, application protocols, and conformance testing within the ISO 10303 series.	2024
IIC	IIC Whitepaper	The IIC Whitepaper provides industry guidance for the development, adoption and use of IIoT standards. The whitepaper presents a visionary strategy to enable interoperability and system compatibility across the entire IIoT ecosystems for the industries.	2021
OI4	Cyber security design principles	This whitepaper defines the Open Industry 4.0 Industrial Cybersecurity framework. The whitepaper provides insights to of key security and privacy principles applicable across the building blocks of the layers of the IIoT ecosystems.	2022
OPC	30081	OPC UA Process Automation Device Information Model – PA-DIM(TM)“ defines a standardized information model for process automation devices in accordance with the OPC Foundation’s OPC-UA specifications	2022
OPC	30080-5	The Information Model for FDI Hosts. One of the main tasks of the Information Model is to reflect the topology of the automation system. Therefore it represents the devices of the automation system as well as the connecting communication networks	2022
ISA	ISA99 / 62443	This category is entitled as "Cybersecurity for industrial automation & control systems". This category includes security technologies and program requirements with concepts and models for industrial automation.	2020
ISA	ISA106	This category is entitled as "Procedure Automation for Continuous Process Operations". This category includes standards for procedure automation models and work processes.	2023
ISA	ISA100	This category is entitled as "Wireless Systems for Industrial Automation". This category includes standards for network management, secure connectivity over untrusted networks, and trustworthiness.	2017
IEC	SS IEC 62443-2-1	Industrial communication networks – Network and system security – Part 2-1: Establishing an industrial automation and control system security program	2018
IEC	TR IEC/TR 62443-2-3	Security for industrial automation and control systems - Part 2-3: Patch management in the IACS environment	2018
IEC	IEC 62443-4-2	Security for industrial automation and control systems - Part 4-2: Technical security requirements for IACS components	2019
IEC	SS IEC 62443-2-4	Security for industrial automation and control systems - Part 2-4: Security program requirements for IACS service providers	2018
IEC	SS IEC 63278-1	This definition proposes Asset Administration Shell (AAS), that standardizes the exchange and mutual use of information between software applications in industrial processes to ensure secure and trusted interactions. It applies across various types of industrial processes and sectors, covering the entire lifecycle of physical, digital, or intangible assets.	2023
IEC	TR IEC/TR 62443-3-1	Industrial communication networks - Network and system security – Part 3-1: Security technologies for industrial automation and control systems	2018

⁵**Note:** We consider the latest update date of the document stated in the SDO repositories. There may be more recent in-progress yet unpublished documents due to ongoing SDO work group efforts.

batch process data exchanged between sensors, instruments, controllers, and software systems. The International Society of Automation (ISA) [156], is a non-profit professional

association established to empower the global automation community through standards and knowledge sharing. ISA focuses on defining standards in various aspects of industrial

automation, including cybersecurity, procedure automation, and wireless systems. Apart from standards development, ISA offers certifications, education, and training programs aimed at enhancing industrial automation processes.

Once each standard developed by different SDOs are being considered, each standard establishes distinguishing aspects on different contexts of industrial applications, including automation and communication. In the research of industrial automation and communication for the Industry 5.0, it is important to precisely identify the applicable industrial standards, and ensure that the research deliverables are aligned with the latest standards defined by each SDO.

B. RESEARCH PROJECTS

Industrial automation is a dynamic research field, capturing considerable attention from industry and academia. Numerous projects, born out of collaborations between industry and academia, are actively exploring the realms of industrial automation and the prospective integration of 5G and beyond networks. The following offers a concise overview of each project, encompassing those recently concluded as well as those currently in progress. Collectively, these initiatives contribute to advancing the understanding and application of advanced communication technologies for industrial automation.

1) APRIL

The APRIL project [157], is dedicated to pioneering research in the development of autonomous, dexterous, and market-oriented robot prototypes. The project's primary objective is to introduce innovative approaches to automate the manufacturing and processing of flexible and deformable materials.

2) SHOP4CF

The SHOP4CF project [158], focuses on achieving a harmonious balance between cost-efficient automation for repetitive tasks and active human engagement in areas such as adaptability, creativity, and agility. The initiative seeks to promote a highly connected and automated factory model, utilizing the vast amount of data generated within manufacturing facilities. SHOP4CF plans to sustain its endeavors through a self-sufficient non-profit association with members spanning across Europe.

3) CONVERGING

The CONVERGING project [159], aims to design, implement, validate, and advocate for intelligent and adaptable production systems. These systems incorporate various autonomous agents, including collaborative robots, AGVs, and human workers, capable of functioning effectively in diverse production settings. The project emphasizes a multi-tiered AI-driven understanding at different levels, enabling collective perception facilitated by the Digital Pipeline. This facilitates harmonious interactions between resources within a "social industrial environment," ensuring a trustworthy, secure, and inclusive user experience.

4) MASTERLY

The MASTERLY project [160], endeavors to address the challenges associated in mass production transformations towards more customized approaches with adaptable robotic solutions. The initiative involves incorporating modular grippers, cutting-edge robotic technologies, including mobile and industrial robots with varying payload capacities, smart cranes, and integrating advanced AI-driven control and perception capabilities for autonomous operation. These solutions aim to autonomously handle diverse parts in terms of size, shape, and material while ensuring universal acceptance among workers of all genders. The effectiveness of these proposed technologies will be evaluated through testing in three distinct use cases across different production sectors, with the goal of showcasing their flexibility, efficiency, and user acceptance while demonstrating applicability, scalability, and adaptability across production lines and sectors.

5) REMODEL

The REMODEL project [161], is envisioned to support the new production environments for manufacturing products with extensive wires and cables. It integrates a dual-arm robot in the production line for advanced routing and fitting. The project has resolved significant electric wire manipulation issues, as well as add new perception and interaction functions with proven results for effectiveness in four industrial cases.

6) 6G FLAGSHIP

6G Flagship [162], is the world's first 6G research programme, which is also part of the Finnish government's national research spearhead programme from 2018 to 2026. The main objective of the 6G Flagship is to create the essential technological 6G components, the tools and the equipment to build a 6G test network, to develop selected vertical applications for 6G to accelerate the digitalization of society and to continue to play a recognised pioneering role in global 6G research.

7) SMART5GRID

Smart5Grid [163], is a pioneering initiative that transforms traditional power distribution grids into advanced and automated smart grids and integrates a state-of-the-art communication network. The project addresses the communication challenges posed by device density, bandwidth and ultra-low latency, which are critical for efficient and secure smart grid operation. Using innovations such as the Smart5Grid MEC platform, 5G Core Service-Based Architecture and Network Slicing, Smart5Grid enables online monitoring, remote maintenance and rapid fault detection within a 20ms power cycle to usher power utilities into a new era of reliability and QoS.

8) ARCADIAN-IOT

ARCADIAN-IoT project [164], funded by the European Union, proposes a framework for greater security, trust

and privacy in IoT systems, while offering users greater control over their privacy with improved transparency. The project envisions autonomous trust, security and privacy management for the IoT.

9) TARGET-X

The TARGET-X initiative [165], aims to accelerate the digital transformation of key industry sectors energy, construction, automotive and manufacturing through extensive trials in multiple testbeds. By presenting and evaluating 5G/6G technologies such as real-time communication, digital twinning and sensor network data fusion in practise, the project focuses on Key Performance Indicators (KPIs) with a special emphasis on Key Value Indicators (KVIs) such as sustainability, security and privacy. The results will feed into the development of new business models and an evaluation framework led by the TARGET-X community and a consortium of SMEs, IT and OT partners and up to 100 Future Talent Programs (FTSP) projects. This will strengthen the 5G/6G ecosystem and contribute to industry automation standards.

10) 6G-SANDBOX

The 6G-SANDBOX initiative [166], funded by the Horizon program (HORIZON-JU-SNS-2022-STREAM-C-01-01) is a comprehensive and modular research project that aims to improve the European experimentation ecosystem for research, including communication for industrial automation. With a focus on supporting technology and research validation processes that are essential for the evolution to 6G in the next decade, the project presents Trial Networks. These networks are fully configurable, manageable and controlled end-to-end systems that include both digital and physical nodes and are distributed across EU locations such as Malaga, Athens, Berlin and Oulu. 6G-SANDBOX not only offers third parties automated experimentation opportunities through an extensive toolbox, but also aims to achieve a sustainable impact by releasing quantified 6G KPIs and KVIs to interested parties and establishing an open repository for sharing developments and Application Programming Interfaces (APIs) beyond the project scope, thus contributing to the formation of a European 6G library.

VII. OPEN CHALLENGES

As industrial automation evolves alongside B5G/6G networks, future systems must not only address existing constraints, but also anticipate emerging needs for intelligent, autonomous and adaptive infrastructures. To this end, our vision focuses on scalable integration, resilient interoperability and proactive security frameworks that enable self-optimising industrial environments. We have divided the open challenges into three subcategories: integration, interoperability, and challenges related to security and privacy.

Architectural Convergence and OT/IT Integration: One of the main complexities in the operation of industrial

automation systems is the integration of the new infrastructure with the legacy systems, which requires high investments and efforts. This is because legacy systems are often outdated and incompatible with newer technologies, making them difficult to integrate into the new infrastructure [167]. In addition, the integration process can be complicated by the presence of multiple legacy systems that may need to be integrated into the new infrastructure. Existing factory setups often rely on deterministic wired systems (e.g., PROFINET, EtherCAT), making it difficult to introduce wireless solutions without compromising reliability or safety certifications. Furthermore, the integration process can require significant time and resources, which can lead to delays and higher costs. The integration process can also be complicated by the presence of multiple legacy systems. To ensure co-existence, future architectures must support hybrid determinism, allowing TSN-over-6G to interface seamlessly with wired time-critical networks. One promising direction is the design of hybrid co-simulation platforms based on digital twins that can be used to test and iteratively refine integration strategies before they are fully deployed. Studies such as the one by [3] have explored modular integration frameworks that enable incremental integration while minimizing system downtime. However, these often require high initial investment and complex project management.

Heterogeneous Protocol Harmonization: Seamless communication across diverse systems is essential for industrial automation to realize its full potential [168]. However, achieving interoperability can be challenging for several reasons: (i) *Lack of standardization:* Many different protocols and standards are used in industrial automation, and without a common standard, it can be difficult for different systems to communicate effectively. (ii) *Legacy systems:* Many industrial automation systems have been in operation for many years and may use outdated technologies that are no longer supported or are incompatible with newer systems. (iii) *Security concerns:* Interoperability can also be challenging due to security concerns, as it requires sharing sensitive data across different systems, which can be difficult to secure. (iv) *Different data formats:* Industrial automation systems may use different data formats and languages, which can make it difficult to exchange data between systems. Research by [169] highlights the use of middleware to convert data between formats as a partial solution, although this introduces additional latency and potential points of failure. (v) *Lack of testing:* interoperability between different systems may not be adequately tested and validated, which can increase the risk of errors and reduce the overall effectiveness.

On the other hand, the use of open standards and protocols such as OPC UA or MQTT can ensure that different systems can communicate with each other. Software adapters can be used to translate messages between different protocols so that different systems can communicate with each other even if they use different protocols. Using common data formats such as XML or JSON can ensure that data can

be shared between different systems [169]. Finally, testing and validating the interoperability of different systems can help ensure that they work together as expected and identify problems that need to be fixed. Note also that technology is one of the key elements in building interoperability in industrial automation. However, one of the biggest challenge is likely not be technological or infrastructural, but the need to build consensus and alignment among players from different industries with different interests [170]. This will require collaboration at scale, defining a clear business case, and building a differentiated value proposition for industrial automation stakeholders. Future systems could leverage AI-powered mediation layers to dynamically harmonize heterogeneous protocols and resolve compatibility conflicts in real-time.

Time-Critical Orchestration and Predictability: While B5G/6G promises uRLLC, ensuring consistent microsecond-level jitter bounds across distributed industrial control systems remains an unresolved problem. Variability from multi-hop wireless, external interference, and cloud-edge orchestration introduces uncertainty that jeopardizes control cycles. The development of predictive orchestration frameworks that utilise AI/ML to anticipate network conditions and pre-assign deterministic slices can provide some solutions. Real-time task offloading to fog and edge nodes could be made self-optimising through feedback from cognitive digital twins embedded in the control loop.

Cybersecurity in converged domains: Potential security risks in the integration of industrial automation with enhanced communication infrastructure include: (i) *DDOS (Distributed Denial of Service) attacks*: Faster speeds enable faster data transfers, which can facilitate fast cyberattacks such as DDOS (Distributed Denial of Service) attacks. (ii) *Malware propagation*: Low latency can make it difficult for security systems (e.g., firewalls, intrusion detection systems) to detect and respond to malicious activity in real time, allowing the malware to spread further and cause more damage before it is detected and stopped. Therefore, it is important to find a balance between the need for low latency and effective security measures to minimize the risk of malware spreading. (iii) *Attack surface*: The proliferation of networked devices (e.g., IIoT) increases the attack surface, as more devices and sensors are connected to the network and generating data. This can make it more difficult to secure the network and protect against cyber attacks. Hence, securing a large number of different devices, each with its own security vulnerabilities, is becoming a challenge. (iv) *Unauthorized access*: Network slicing requires sophisticated security measures to prevent unauthorized access between slices and to protect sensitive data. For instance, recent advances in network slice isolation techniques have been proposed to secure data transit between slices, yet these require complex configuration and continuous updates to stay ahead of potential threats [171]. If a user gains access to a network slice that they are not authorized to use, this

can lead to unauthorized access. One way this can happen is through insufficient authentication and authorization. If a user is not properly verified and granted access to a network slice, they can access resources and data that they should not have access to. Another possibility of unauthorized access is a vulnerability in the network slice. If there is a security vulnerability in the network slice, an attacker can exploit it to gain access to the slice and the resources and data it contains. To prevent unauthorized access, it is important to implement strong authentication and authorization policies and to regularly monitor and update the network slice to ensure it is protected from potential vulnerabilities. (v) *Edge device security*: Edge computing decentralizes data processing, so it is important to secure edge devices and ensure the integrity and confidentiality of data at the edge. In addition to the increased attack surface, edge computing can also create new vulnerabilities, such as weak or easy-to-guess passwords on devices or unsecured communication protocols. Proper device configuration, network segmentation and security protocols can help minimise these risks. (vi) *Security of data at rest and in motion*: Image, container, network and platform security all play a crucial role in securing data at rest and in motion. When creating images, it is important to use secure settings and configurations to prevent unauthorized access. This includes using strong passwords, enabling multi-factor authentication and configuring access controls. The introduction of zero-trust architectures at OT level is essential. Solutions such as self-sovereign identity mechanisms using decentralized ledgers for device authentication and lightweight on-device anomaly detectors for real-time threat response without cloud dependency can be promising. 6G-native security signaling and dynamic policy propagation can also be explored. To further strengthen defences, federated threat intelligence sharing between industry clusters could be used to jointly detect and respond to advanced persistent threats.

Intelligence at the Edge for Cognitive Automation: Cognitive digital twins, anomaly detection and adaptive motion control require real-time inference at the edge. However, deploying large AI models in constrained industrial devices poses challenges in terms of latency, energy and trustworthiness of outputs. A hierarchical, AI-native continuum, where low-level reactive tasks (e.g., vibration monitoring) can be handled by ultra-compact models at the edge, while complex spatio-temporal reasoning (e.g., process optimization) is passed to micro data centres via B5G/6G, may provide a viable solution. Explainable AI modules and certified runtime environments may be necessary to ensure operational trust in such deployments.

In conclusion, the above challenges not only emphasise the complexity of evolving industrial networks, but also provide a roadmap for future innovation. Addressing them requires cross-disciplinary collaboration, proactive standardisation efforts and adaptable technologies that can evolve with the demands of Industry 5.0.

VIII. REAL-WORLD TECHNOLOGY IMPLEMENTATIONS IN INDUSTRY 4.0/5.0

A. AUTOMOTIVE MANUFACTURING

To improve factory automation and quality control, major automakers have started implementing private 5G networks. For instance, in partnership with Ericsson and AWS, Hitachi Astemo's EV-motor plant in Kentucky installed an on-site 5G/edge system that streams 4K assembly-line video to AI analytics. Due to 5G's high throughput and low latency, trials showed that 24 motor components could be inspected simultaneously (as opposed to one at a time) with sub-millimeter defect detection. Rapid deployment for Industry 4.0 use cases was demonstrated by the 5G wireless installation, which was finished in a few days. In a similar vein, Audi and Ericsson demonstrated in the lab how 5G URLLC can take the place of wired robot cells [172]. The system achieved safety and flexibility that Wi-Fi cannot match by instantly stopping a robot arm whenever a human entered its workspace.

B. STEEL AND INDUSTRIAL MANUFACTURING

Steel Plants: Steel manufacturers have been at the forefront of utilising private 5G to connect heavy machinery and improve safety and efficiency. ArcelorMittal France's "5G Steel" programme (with Orange Business and Ericsson) built one of the world's largest industrial 4G/5G networks in its Dunkirk and Mardyck plants. On the hot-strip mill, 5G-connected mobile tablets substitute paper for inspection and lockout/tagout procedures, and AR headsets by the operators give access to drawings or real-time expert support through the network. In the scrapyard, 5G connects weight and composition of materials sensors with crane operators, and they receive real-time quality data from the production system directly. The network uses slicing [173] to assign high-performance links to mission-critical applications (e.g., autonomous vehicle control) and isolate traffic. Planned applications in the future are autonomous railroad shuttles on the plant 5G network and wireless AGVs in the yard, aimed at reducing downtime and enhancing safety. A standalone Siemens/Salzgitter trial in Germany also used a commercial 5G (3.7 GHz) over a flat-steel line at a steel factory [174]; the initial application was to control AGVs carrying red-hot slabs remotely, which required robust, low-latency connections in a harsh, metal-shielded environment.

Discrete Industrial Manufacturing: 5G pilots are also being rolled out in wider factory settings. For instance, China's Haier worked with Huawei and China Mobile to deploy 5G-capable edge computing and AI vision at a refrigerator plant. A high-power industrial camera installed atop a robotic arm checks each stainless-steel door for scratches; the 5G-linked local edge server processes images in real time so faulty units can be rerouted without stopping the line. Worcester Bosch (UK boiler-for-home manufacturer) launched a 5G smart factory (Ericsson/BT) in the U.K. where real-time 5G sensor network machine data is supplied to predictive maintenance analytics. Early returns saw $\approx 2\%$

higher throughput after optimized equipment availability, and a novel 5G collision-detection sensor grid significantly improved workers' safety. These illustrate how 5G and edge computing together can give real-time sensor and camera feedback (to enable online quality inspection and prediction of failure), thus reducing unplanned stops and waste. In fact, there's been cited cognitive digital-twin use in a steel-pipe plant which utilized AI models on real-time streamed sensor data to forecast faults and schedule maintenance; it saved $\approx 10\%$ energy and removed the majority of unplanned stops.

C. PHARMACEUTICAL MANUFACTURING

In life-sciences applications, smooth connectivity is critical. In a recent pilot in the Bio-Manguinhos vaccine plant (Fiocruz, Brazil), a private Nokia 5G network was used to monitor rolling incubators in a series of vaccine API batches remotely [175]. Wireless coverage dead spots (due to metal fermenter rooms) previously caused sporadic batch interruption and possible waste of over two million doses. The new 5G link (≈ 16 ms latency) punched through the steel shielding and streamed data in real time from PLCs and accelerometers on the roller tables. The plant engineers report that the 5G installation "quickly resolved" the disruption issue: it enabled real-time IoT dashboards and even a smartphone app for maintenance crews, remote monitoring of roller performance. This floor test case is said to have avoided reagent waste and enhanced yield in critical vaccine production. In the same manner, Bayer's Turku (Finland) pharma plant has partnered with Verizon to create a 5G innovation center. Though it is at pilot stage currently, it is in the areas of such applications as AR-augmented machine maintenance and digital tracking of products along the line [176]. These solutions demonstrate how 5G will strengthen batch traceability and enable regulatory control: for example, real-time connectivity enables automated logging of process parameters and allows rapid remote intervention to avoid costly downtime.

IX. FUTURE OUTLOOK (6G AND INDUSTRY 5.0)

So far, all deployments likely use 5G (private 4G/5G or standalone 5G). Commercial 6G systems are not yet on the horizon but research consortia such as 6G-IA, ETSI, already outline new use cases for the 2030s. Potentially offered 6G features – ultra-high bandwidth, sub-millisecond latency (μ s-scale) and integrated sensing/AI – hold the promise of even closer cyber-physical integration in future factories. For instance, 6G is expected to make possible real-time holographic telepresence (essentially immersive augmented reality), large digital twins that learn and adapt at device speeds, and networked AI inference. In the meantime, the current 5G trials have already delivered measurable dividends. Companies testify vision inspection across networks and AI lowered defect rates and increased quality assurance; predictive maintenance software on 5G is sensing failure prior to occurrence (improving overall equipment effectiveness); and mobile robots and workers are

able to move more safely and easily under 5G control. These case studies, taken together across automotive, steel, manufacturing and pharmaceuticals, demonstrate in combination that 5G (usually complemented with edge computing, AI, AR or digital twins) is already improving uptime, quality and sustainability in real factories, on the way to the human-centric and resilient goals of Industry 5.0

X. LESSONS LEARNED AND FUTURE DIRECTIONS

The following subsections explain the lessons learned, some of the open challenges in industrial automation and future research directions in terms of architecture, security and efficiency.

A. LESSONS LEARNED

6G should be able to support different deployment options to meet the requirements of industrial applications accordingly. Therefore, to evaluate and determine the actual need for connectivity, it is important to understand the industrial application. Some of the lessons we have learned in this study are listed below:

Capabilities of connection: Wireless connectivity may not have the same capabilities as wired technology. In some cases, connectivity requirements are higher than what the application really needs. In a pick-and-place task, for example, where a robot is required to pick up objects and place them in a different location, the connectivity, reliability and latency requirements are less stringent than for a robotic arm that welds or paints, which requires more complicated and precise operations to avoid defective products. Some other additional techniques such as the use of a high-resolution camera to detect and correct misalignments or inconsistencies, a servo motor for precise control, a feedback sensor that provides real-time feedback on the position and movement of the arm, the use of a redundant system that ensures that the task can be completed even if a component fails, a closed-loop system that ensures the arm is always in the correct position and can make real-time adjustments to ensure accurate placement, or a high-speed conveyor belt can all help to increase the speed of the pick-and-place task. In these scenarios, the use of ML and predictive analytics can be leveraged to adjust connectivity parameters in real-time based on operational complexity and environmental factors, optimizing the use of radio resources in demanding industrial use cases of 6G networks.

Industrial organizations are prioritizing industry automation related use cases versus applications: Many organizations are prioritizing industry automation related use cases before targeting vertical-facing applications. The main reason is that industry automation can help improve efficiency, productivity, and safety within the organization, which can lead to cost savings and competitive advantages. In this context, to promote more industrial applications, network slicing can help create dedicated virtual networks that meet the specific requirements of different industrial automation applications and improve their flexibility and resource utilization.

The introduction of industrial automation brings with it considerable operational complexity: Industry faces significant operational difficulties in implementing industrial automation. As a result, the preferred way of deploying industrial automation is through a phased approach, such as piloting the automation in a limited scope before rolling it out more widely. This helps eliminate the operational difficulties that industries may face in adopting industrial automation, as they can test the automation in a controlled environment and make adjustments as needed before rolling it out on a larger scale. This approach also helps build acceptance and support for automation among employees and other stakeholders, and ensure that automation is aligned with the specific needs and requirements of the industry. In this way, the application of machine learning algorithms can also be used to analyze the operational data from the pilot project and optimize the processes prior to full-scale deployment.

Interoperability and cross-platform integration are critical as it allows different devices and systems to work together seamlessly. A common architecture for interoperability and cross-platform integration must be achieved through the use of open standards and protocols. This will ensure that different devices and systems can communicate with each other regardless of the manufacturer or the technology used. Some examples of open standards and protocols used in industrial automation include Modbus, OPC UA, and MQTT. The applications of interoperability and cross-platform integration are also very extensive in industrial automation and must be examined in detail. One example is a system that enables the integration of different types of sensors, such as temperature and pressure sensors, in a single application [177]. Another example is a system that enables the integration of different control systems, such as PLC and Distributed Control System (DCS), in a single process [178]. Machine learning can further improve this interoperability by providing intelligent translation layers that adapt in real time to the data formats and protocols of the interacting systems.

Importance of a strong community and industry engagement: The involvement of various stakeholders from the industrial automation community, including manufacturers, communication experts and end users, will lead to the identification of common solutions. Community involvement enables a deeper understanding of the real challenges and contributes to the development of more practical and effective communication strategies [179]. Active collaboration with industry professionals and end users can provide valuable insights into the specific communication needs in industrial automation. Collaboration with industry partners can enable real-world testing to validate the effectiveness of communication strategies in different industrial environments [180].

Development of new revenue streams and improvement of industrial client retention: When it comes to using mobile communication for industrial automation, it will be important to tailor mobile communication services to

the specific requirements of industrial clients or verticals. Customized offerings, such as value-added services, will adapt these services to the unique requirements of industrial automation and thus contribute to higher vertical satisfaction and loyalty. Another important point is the need to ensure high reliability and redundancy in mobile communication networks. Industries that depend on seamless and reliable communication for their automation processes require robust networks that minimize downtime and thus promote the trust and loyalty of industrial users. Establishing mechanisms to continuously monitor and optimise service performance in the industrial environment will also ensure efficient communication and positive vertical experiences. Network slicing can offer important benefits such as the ability to customise network features to meet specific requirements, increase security and improve operational efficiency.

Industry shifts from short-term monetization to long-term value creation: This is reflected in the increasing focus on developing sustainable and future-proof solutions and on providing comprehensive services and support to verticals. Companies are realizing that short-term monetization may not be enough to maintain vertical loyalty and market position over the long term, and that investing in long-term value creation can lead to greater success in the long run. This change is driven by technological advances, changing vertical expectations and increasing competition in the market. Through the use of advanced technologies such as AI and ML, industrial automation can enable companies to offer personalized and customized products to verticals. This can also lead to higher vertical satisfaction and stronger brand loyalty.

Small-scale industrial automation pilots have been launched to test use cases and define the real needs of industry: A small-scale pilot can test the technology and identify any issues before scaling up. Some small-scale industrial automation pilots have been launched in the industry on various domains including (i) Testing automation in a specific factory or production line to identify the most efficient and effective processes [181]. (ii) Implementing IoT devices to collect data and analyze the performance of machines and equipment [182], [183]. (iii) Using AR and VR to improve training and safety in hazardous environments [184]. (iv) Automating repetitive tasks, such as quality control inspections, to free up employees for more complex and value-added work [185]. (v) Implementing cobots, or collaborative robots, to work alongside human employees and assist with tasks that are too dangerous or difficult for humans [186]. (vi) Using machine learning algorithms to optimize production processes and improve efficiency.

These pilots are crucial to test the effectiveness of network slicing in different industrial scenarios. They enable adjustments based on real-time data processed by ML algorithms to optimize performance and resource allocation. They can help to identify the most promising applications of industrial automation and provided valuable insights into the

real needs of industry. However, large-scale pilots require significant resources and investment, so companies may be hesitant to embark on such a large-scale project without first testing it with a smaller pilot. In addition, large-scale pilots are less manageable and difficult to implement, allowing companies to test new technologies and processes without risking significant disruption to their operations, as is the case with smaller pilots. Finally, companies have shared their experiences and lessons learned from small pilot projects rather than large ones. The reason could be that they may not be willing other companies to learn from their peers and make informed decisions about implementing automation in their own operations in large-scale pilots.

Wired and wireless communication needs: The requirements of industrial applications must be fully understood in order to assess and determine actual needs. For example, a wireless connection may not provide the same functionality as a wired technology, but under many conditions, existing connectivity with wireless technology would already meet the requirements of the industrial application. For instance, when a robot performs a pick-and-place task, the stringent requirements of low latency of few milliseconds and reliability can be relaxed compared to a robotic arm performing welding or painting tasks, the faulty execution of which can result in a defective product [22]. This can increase the number of use cases supported with 6G and Edge/Cloud Continuum for reliable execution, automatic scaling and resource optimization while saving valuable radio resources. The strategic use of network slicing can meet the diverse needs of various industrial automation tasks, from low latency and high reliability requirements for critical operations to less demanding monitoring tasks.

Features selection of wireless connectivity: Depending on the application requirements, better optimizations and interfaces need to be designed between actuator, sensor, industrial application and field-bus. Significant bandwidth requirements in the uplink or changing the network selection during the execution of robotic tasks are some of the requirements that may arise [21]. For example, to fulfil some of them, the wireless packet transmission method can work proactively to detect when a robust transmission is required to avoid fieldbus link failures (which can abruptly stop the robot) in a robotics scenario. At the same time, the radio scheduler can also know the details of the industry protocol, e.g., if the limit for the waiting time of the industry control frame is known, the scheduler can discard the frame if the limit is exceeded. Some optimization/selection of packets to be transmitted in the uplink (for scenarios where many sensors, cameras, are attached to actuators such as robot arms or Automated Guided Vehicles (AGVs)) can help to reduce the amount of messages exchanged [14]. In this context, effective network slicing can take into account the specific connectivity requirements of industrial applications, which can vary significantly even within a single facility. Machine learning algorithms can also be used to predict

TABLE 10. Summary related works on lessons learnt, challenges and future directions.

Research	Summary
Lessons Learned	— Presents autonomous, intelligent and adaptable designs for production systems. [157]–[159] — Investigates the next generation mobile technologies for the industrial automation. [162], [165], [166] — Focuses on secure, reliable and robust frameworks by considering all interacting entities. [163], [164] — Handle objects for the manipulation of objects in new production environments. [160], [161]
Challenges	— Provide inter-connections for different industrial services over multiple wireless technologies. [167], [170] — Manage and operate a network comprising equipment from multiple vendors/technologies. [168], [169] — Identify existing threats and protect manufacturing data against intrusions from outside. [83], [187]–[189] — Ensure a non-disruptive smart factory workflow, and secure business-critical information. [2], [9]
Future Directions	— Enhance industrial systems and devices by providing applications with immutable storage. [8], [91], [191] — Gives devices ownership and control of their digital identities without a central authority. [191], [192] — Develop lightweight cryptographic systems that are secure against quantum computers. [193]–[196] — Serve devices in factory environments with low cost and the use of the lighting infrastructure. [197]–[199]

and dynamically allocate network resources to optimize performance and efficiency based on ongoing operations.

B. FUTURE RESEARCH DIRECTIONS

Although automation in industry is a developing trend, it will probably be some time before it becomes more widespread and is adopted by a larger community of partner companies. Several factors contribute to this outcome, including the cost of implementing automation solutions, the need for expertise and knowledge, and the need for ongoing maintenance and support. In addition, the cultural and organizational changes required to fully adopt automation can also slow adoption. To accelerate the adoption of automation, the development of more cost-effective and user-friendly automation solutions could help speed up the adoption process. Table 10 summarizes the research activities and related works in mobile communication for industry automation.

Future efforts could focus on seamlessly integrating Blockchain Network (BCN)-based Self Sovereign Identity (SSI) solutions with 6G networks for industrial automation [191]. This integration would utilise the advanced capabilities of 6G, such as the increased speed, low latency and increased connectivity of devices, to further improve the security and efficiency of mobile communications in industrial automation [192], [200]. The continuous development of blockchain technology and cryptographic hashing techniques can contribute to the improvement of security measures in SSI systems. Future directions could explore novel cryptographic hashing methods, zero-knowledge proofs, and quantum-resistant algorithms to strengthen the security of SSI in mobile communications.

The evolution of quantum computers has a significant impact on the security of mobile communication networks in industrial systems. The advancement of quantum computers makes traditional cryptographic methods vulnerable, posing risks to the confidentiality and integrity of data exchanged in industrial systems. The post-quantum cryptographic techniques, which are resistant to attacks by quantum computers, have significant implications for industrial automation architecture and applications to ensure robustness against the

vulnerabilities emerging from quantum technology evolution [194]. Suppliers should aim for cryptographic agility in their solutions given the extended lifespan of industrial automation equipment [195]. In addition, the automated systems control critical infrastructure, and any compromise in communication security can have severe consequences. Post-quantum cryptography safeguards against unauthorized access, manipulation of data, and potential disruptions to industrial processes, mitigating operational continuity and safety risks. The design process should be organized according to the protocols that will be employed. As a result, devices can establish a safe processing environment and execute post-quantum protocols. The choices made during the design's feasibility study have a significant impact on the final product. These choices include the algorithm, the computation technique, the power consumption analysis, the device's resilience to side-channel attacks, the encryption/decryption response time, and other factors that are assessed in light of the network use cases [196].

Visible Light Communication (VLC) holds remarkable potential for transforming industrial automation architecture and applications. By utilising the visible light spectrum, VLC offers high data rates and enables fast and efficient communication, which is essential for automated real-time monitoring and control in industrial systems [197]. VLC is robust against electromagnetic interference and ensures reliable and interference-free communication, especially in environments with high electromagnetic noise. VLC's inherent security, where visible light is limited to certain areas, reduces the risk of unauthorized access and data interception. In addition, the integration of VLC into the LED lighting infrastructure not only improves energy efficiency, but also enables precise localization and positioning indoors, contributing to optimized logistics and increased safety in industrial facilities [198]. However, challenges such as the limited range and susceptibility to obstacles still need to be overcome. Future research in VLC for industrial automation should focus on increasing communication range, improving robustness in challenging environments and developing standardized protocols to ensure interoperability and

wide adoption in various industrial applications [199]. This ongoing research shall play a crucial role in unlocking the full potential of VLC for industrial automation. A summary of this section is presented in Table 10.

XI. CONCLUSION

This paper provides a comprehensive examination of the evolving landscape from Industry 4.0 to the emerging stages of Industry 5.0, emphasizing the central role of advanced communication technologies, including 5G and the emerging 6G. The comprehensive analysis of industrial architecture, industrial devices and various industrial applications highlights that the B5G/6G communication systems have the potential to significantly influence specific domains within industrial automation particularly those involving real-time sensing, closed-loop control, and data-driven coordination by enhancing wireless reliability, responsiveness, and scalability.

By examining the experiences and challenges encountered, valuable insights were gained that emphasize the importance of addressing issues related to industrial device design and implementation, system architecture and deployment options. The paper also highlights the dynamic area of machine learning applications and illustrates its role in anomaly detection, troubleshooting support and innovative solutions such as cavity filter tuning with AI. In addition, an examination of SDOs, ongoing research projects and initiatives such as the 6G Flagship project shows the collaborative efforts that are shaping standards and progress in this area.

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